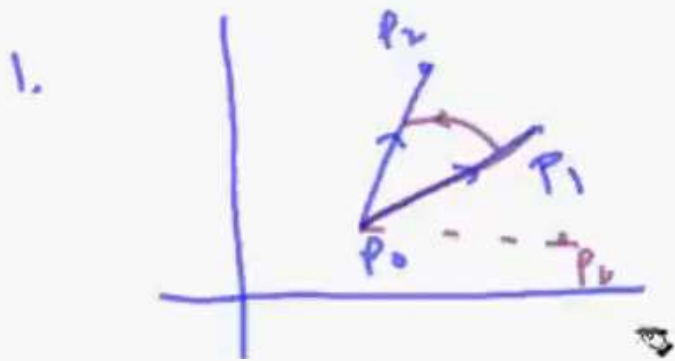


**How to check if two  
given line segments  
intersect?**

we are going to study three problems



IS  $\overrightarrow{P_0P_1}$  CLOCKWISE OR  
ANTI CLOCK WISE FROM  $\overrightarrow{P_0P_2}$

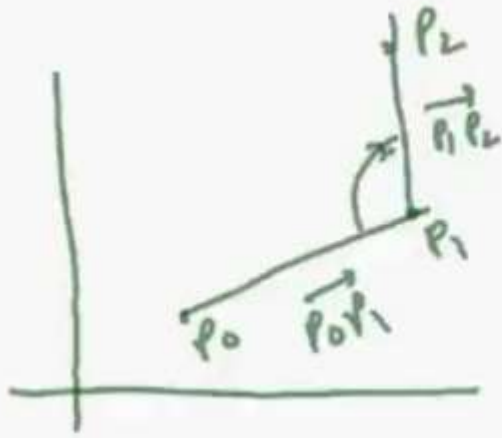
$P_0P_1$  is one segment and  $P_0P_2$  is another segment

Find that  $P_0P_2$ , Does it move counterclockwise respect to  $P_0P_1$ ?

If I have to reach from  $P_0P_1$  to  $P_0P_2$  which is nearer? counterclockwise direction or clockwise (which is larger)

If  $P_0P_2$  is at the bottom(See the figure) then, Clockwise is nearer.

2.

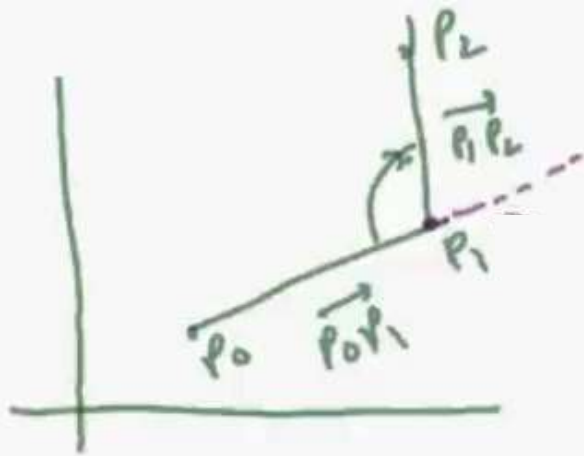


IF WE TRAVERSE  
FROM  $\vec{P_0P_1}$  TO  $\vec{P_1P_2}$   
DO WE MAKE LEFT OR RIGHT  
TURN

$P_0P_1$  is one segment and  $P_0P_2$  is another segment

When we go from  $P_0$  to  $P_1$  and then to  $P_2$ , Do we make a LEFT turn or RIGHT turn?

2.



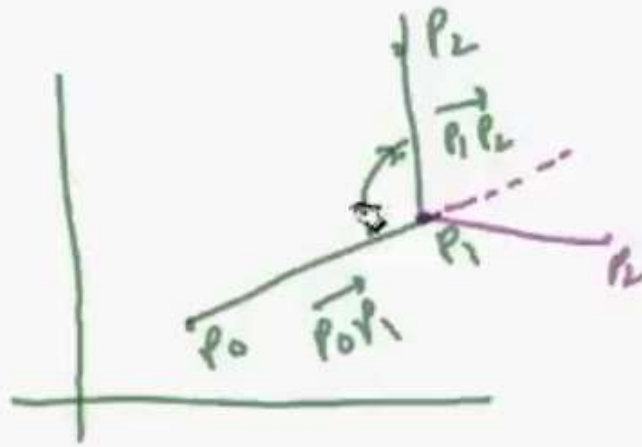
IF WE TRAVERSE  
FROM  $\overrightarrow{P_0P_1}$  TO  $\overrightarrow{P_1P_2}$   
DO WE MAKE LEFT OR RIGHT  
TURN

$P_0P_1$  is one segment and  $P_0P_2$  is another segment

When we go from  $P_0$  to  $P_1$  and then to  $P_2$ , Do we make a LEFT turn or RIGHT turn?

Extend  $P_0P_1$  as shown above and thus we are making a LEFT turn to go to  $P_1P_2$

2.



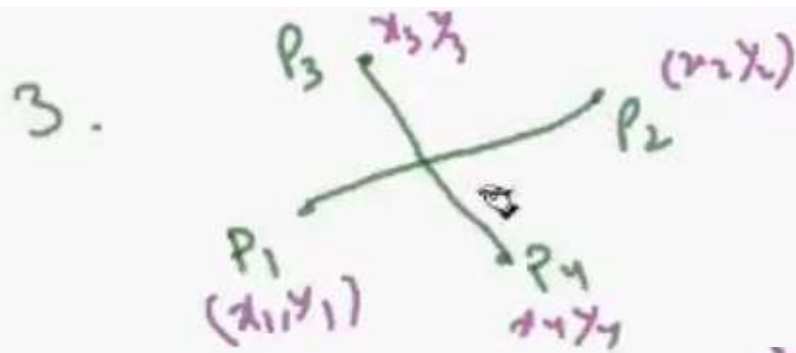
IF WE TRAVERSE  
FROM  $\overrightarrow{P_0P_1}$  TO  $\overrightarrow{P_1P_2}$   
DO WE MAKE LEFT OR RIGHT  
TURN

$P_0P_1$  is one segment and  $P_0P_2$  is another segment

When we go from  $P_0$  to  $P_1$  and then to  $P_2$ , Do we make a LEFT turn or RIGHT turn?

Extend  $P_0P_1$  as shown above and thus we are making a LEFT turn to go to  $P_1P_2$

If  $P_1P_2$  is at the bottom(See the figure) then, it would have been a RIGHT turn



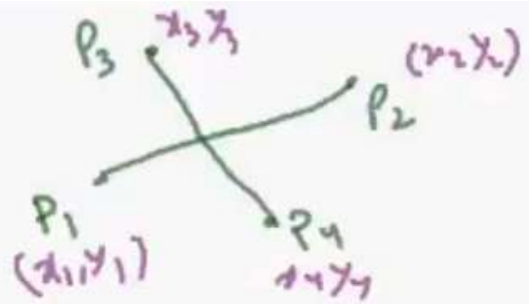
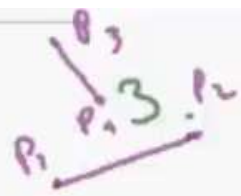
DO SEGMENTS  $\overline{P_1P_2}$   
AND  $\overline{P_3P_4}$  INTERSECT



These 2 line segments do  
not intersect

$P_1P_2$  is one segment and  $P_3P_4$  is another segment

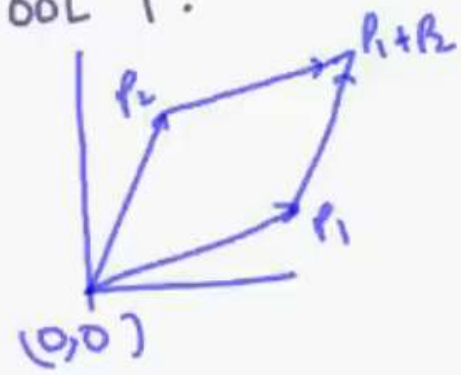
Find Do they intersect? (as per 2 examples given above)



DO SEGMENTS  $\overrightarrow{P_1 P_2}$   
AND  $\overrightarrow{P_3 P_4}$  INTERSECT

TO ANSWER THESE QUESTIONS LET'S  
DEVELOP SOME TOOLS

TOOL 1.



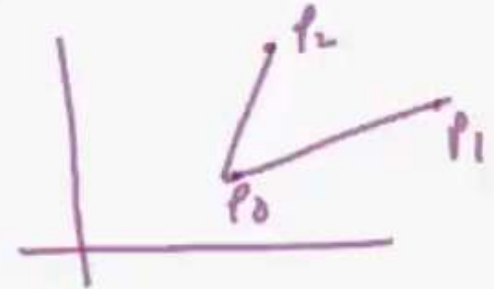
A VECTOR  $\overrightarrow{P_0 P}$  WITH  $P_0 = (0,0)$   
CAN BE WRITTEN AS  $\vec{P}_i$

CROSS PRODUCT

$$\vec{P}_1 \times \vec{P}_2 = \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} = x_1 y_2 - y_1 x_2 = \vec{P}_2 \times \vec{P}_1$$

IF  $\vec{P}_1 \times \vec{P}_2$  IS POSITIVE  
 $\Rightarrow \vec{P}_1$  IS CLOCK WISE FROM  $\vec{P}_2$   
 IF  $\vec{P}_1 \times \vec{P}_2 < 0$  COUNTER CLOCKWISE

FOR LINE SEGMENTS  $\overrightarrow{P_0 P_1}$  AND  $\overrightarrow{P_0 P_2}$   
TO FIND IF  $\overrightarrow{P_0 P_1}$  IS CLOCK WISE / COUNTER CLOCKWISE  
SHIFT  $P_0$  TO BE ORIGIN



i.e. We have to shift the origin

$P_1' = P_1 - P_0$  (i.e.  $P_1$  in the new origin)

$P_2' = P_2 - P_0$  (i.e.  $P_2$  in the new origin)

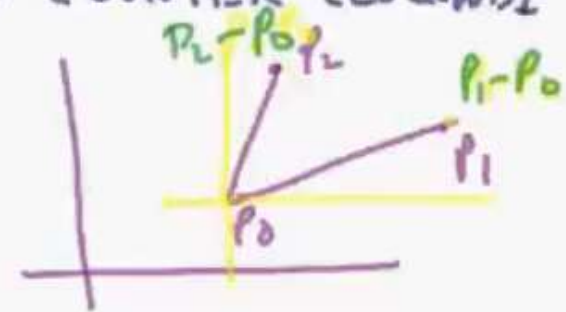


FOR LINE SEGMENTS  $\overrightarrow{P_0 P_1}$  AND  $\overrightarrow{P_0 P_2}$   
 TO FIND IF  $\overrightarrow{P_0 P_1}$  IS CLOCK WISE / COUNTER CLOCKWISE  
 SHIFT  $P_0$  TO BE ORIGIN

$$\Rightarrow \vec{P_1'} = (P_1 - P_0) \quad \vec{P_2'} = (P_2 - P_0)$$

$$\vec{P_1'} \times \vec{P_2'} = (P_1 - P_0) \times (P_2 - P_0)$$

$$= \begin{vmatrix} x_1 - x_0 & x_2 - x_0 \\ y_1 - y_0 & y_2 - y_0 \end{vmatrix} = (x_1 - x_0)(y_2 - y_0) - (x_2 - x_0)(y_1 - y_0)$$



IF  $\vec{P_1'} \times \vec{P_2'} > 0$   $\overrightarrow{P_0 P_1}$  IS CLOCKWISE FROM  
 $\overrightarrow{P_0 P_2}$  ELSE COUNTER CLOCKWISE

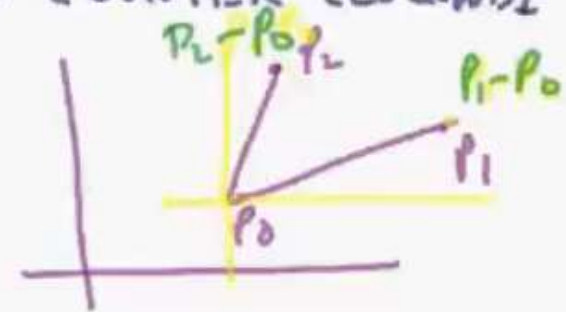
**Note:** basically you will get the **direction** by calculating this cross product

FOR LINE SEGMENTS  $\overrightarrow{P_0 P_1}$  AND  $\overrightarrow{P_0 P_2}$   
 TO FIND IF  $\overrightarrow{P_0 P_1}$  IS CLOCK WISE / COUNTER CLOCKWISE  
 SHIFT  $P_0$  TO BE ORIGIN

$$\Rightarrow \vec{P_1'} = (P_1 - P_0) \quad \vec{P_2'} = (P_2 - P_0)$$

$$\vec{P_1'} \times \vec{P_2'} = (P_1 - P_0) \times (P_2 - P_0)$$

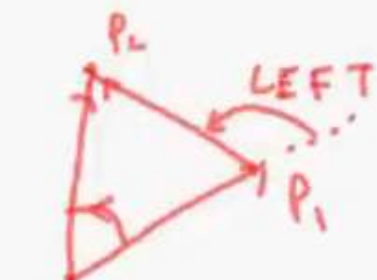
$$= \begin{vmatrix} x_1 - x_0 & y_1 - y_0 \\ x_2 - x_0 & y_2 - y_0 \end{vmatrix} = (x_1 - x_0)(y_2 - y_0) - (x_2 - x_0)(y_1 - y_0)$$



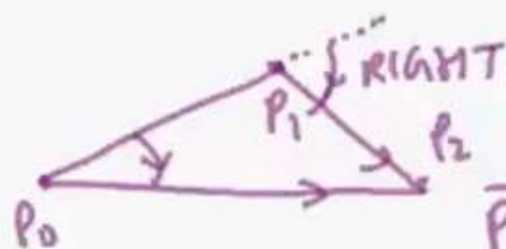
IF  $\vec{P_1'} \times \vec{P_2'} > 0$   $\overrightarrow{P_0 P_1}$  IS CLOCKWISE FROM  
 $\overrightarrow{P_0 P_2}$  ELSE COUNTER CLOCKWISE

have two line segments starting at the same origin  
 so when we have to move towards other line, check  
 do we have to move clockwise or counterclockwise ?

ANS 2.



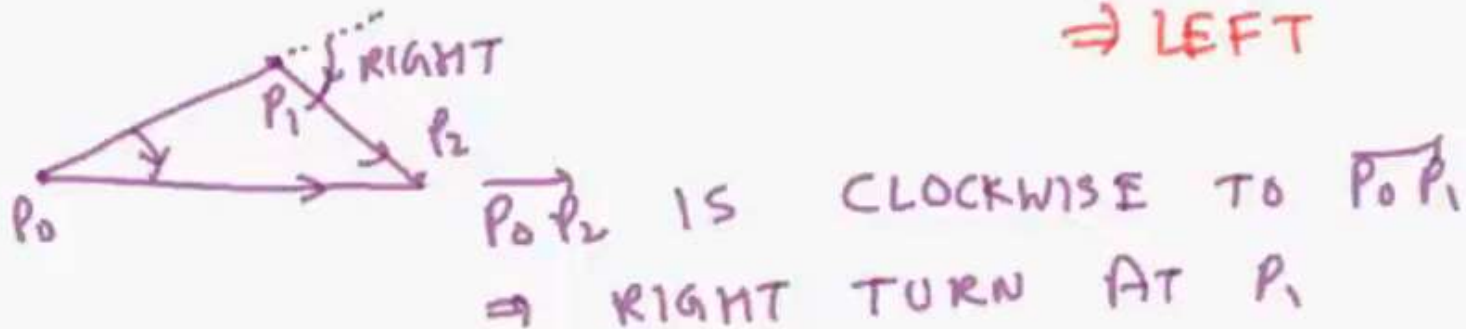
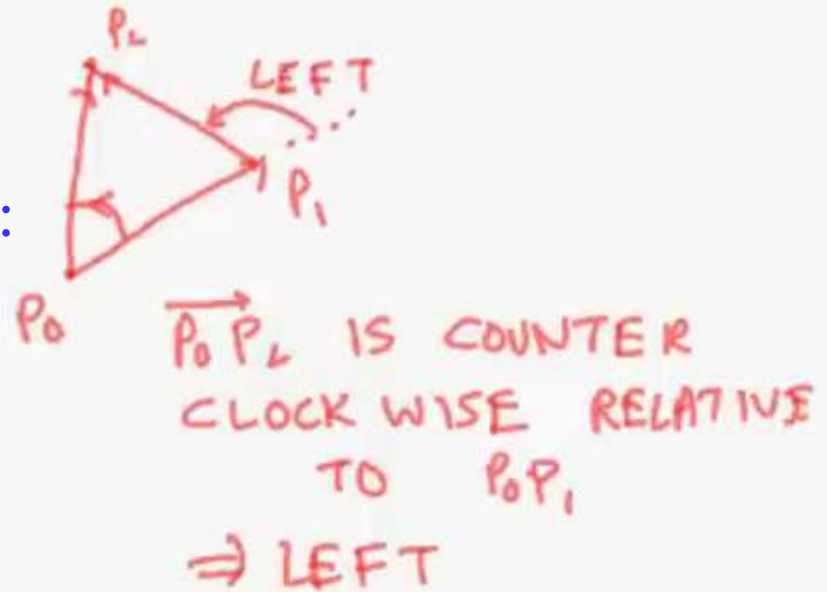
$\vec{P_0 P_2}$  IS COUNTER  
CLOCK WISE RELATIVE  
TO  $\vec{P_0 P_1}$   
 $\Rightarrow$  LEFT



$\vec{P_0 P_2}$  IS CLOCKWISE TO  $\vec{P_0 P_1}$   
 $\Rightarrow$  RIGHT TURN AT  $P_1$

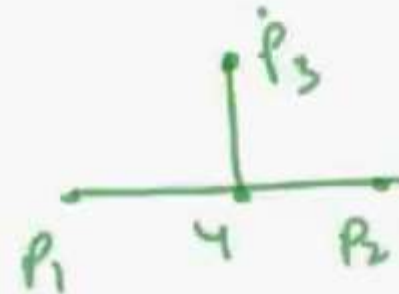
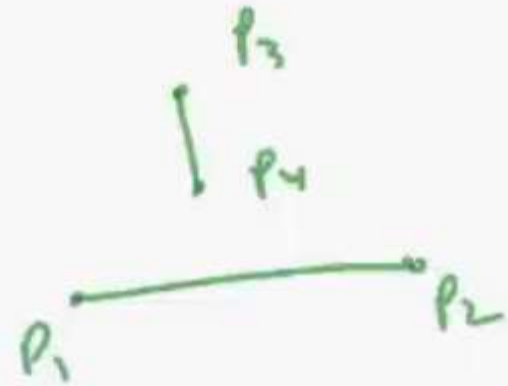
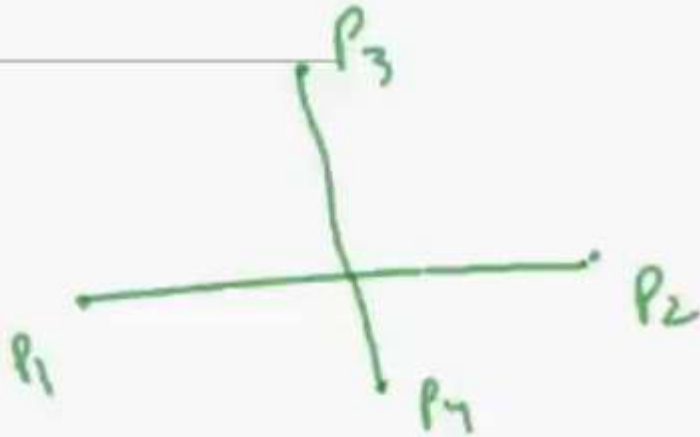


From  $P_0P_1$  to  $P_0P_2$ :  
counterclockwise

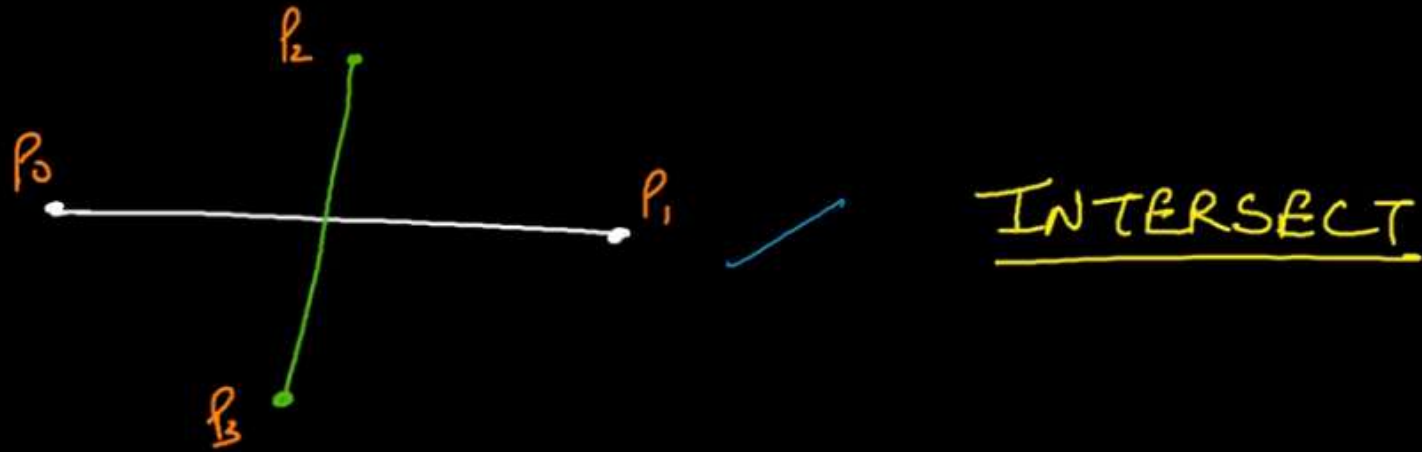


From  $P_0P_1$  to  $P_0P_2$ : clockwise

### 3. Check if two line segments intersect



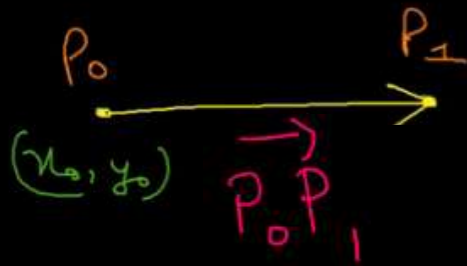
# Check if two line segments intersect



Before understanding the solution to this problem, we need some concept on vectors

# Check if two line segments intersect

CONCEPT OF VECTOR :-



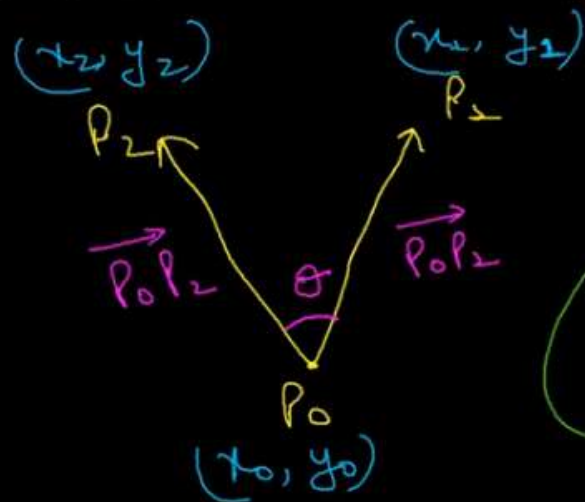
if  $P_0$  is origin  
then

$$\vec{P_0P_1} = P_1 - P_0 = (x_1 - x_0, y_1 - y_0)$$

Drawn a vector  $\vec{P_0P_1}$ , so  
if you are given two points  $P_0P_1$  then we can draw vectors, let's  $\vec{P_0P_1}$  and  
it is called  $\vec{P_0P_1}$  and the corresponding coordinates have also been mentioned.  
If  $P_0$  is Origin,  $\vec{P_0P_1}$  will be coordinates of  $P_1$



# CONCEPT OF CROSS PRODUCT :- (2D) RH-Rule



$$\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta$$

$$\vec{P_0P_1} \times \vec{P_0P_2} = \begin{vmatrix} x_1 - x_0 & y_1 - y_0 \\ x_2 - x_0 & y_2 - y_0 \end{vmatrix}$$

if  $P_0 = \text{Origin}$ ,

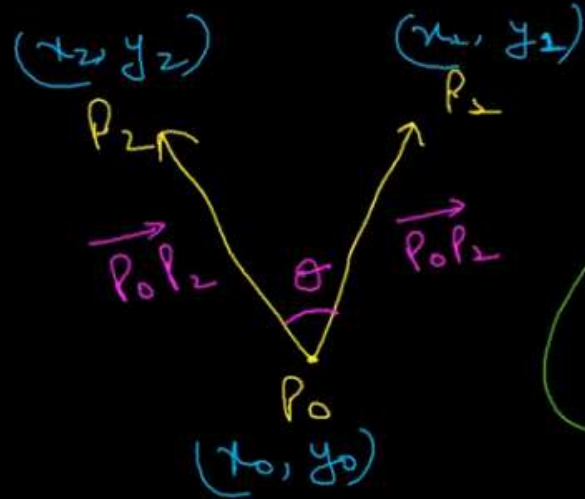
Then.

$$\vec{P_0P_1} \times \vec{P_0P_2} = \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}$$

- 1) Consider 2 vectors  $\vec{P_0P_1}$  and  $\vec{P_0P_2}$  with the given coordinates and angle between them is  $\theta$
- 2) Find the 2 vectors  $\vec{P_0P_1}$  and  $\vec{P_0P_2}$
- 3) Find the cross product. i.e. Find the determinant



# CONCEPT OF CROSS PRODUCT :- (2D) RH-Rule



$$\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta$$

$$\vec{P_0P_1} \times \vec{P_0P_2} = \begin{vmatrix} x_1 - x_0 & y_1 - y_0 \\ x_2 - x_0 & y_2 - y_0 \end{vmatrix}$$

if  $P_0 = \text{Origin}$ ,

then

$$\vec{P_0P_1} \times \vec{P_0P_2} = \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}$$

RH rule – Right Hand Rule.

Goal : to find the direction

i.e. if we multiply 2 vectors, whether it will be positive or negative.

$P_0P_1$  – as index finger and  $P_0P_2$  as middle finger, then Direction of thumb denote the value will be positive or negative. **Tell, For this Case, whether it is positive or negative?**

# CONCEPT OF CROSS PRODUCT :- (2D) (RH-Rule)

$(x_2, y_2)$

$P_2$

$(x_1, y_1)$

$P_1$

$$\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta$$

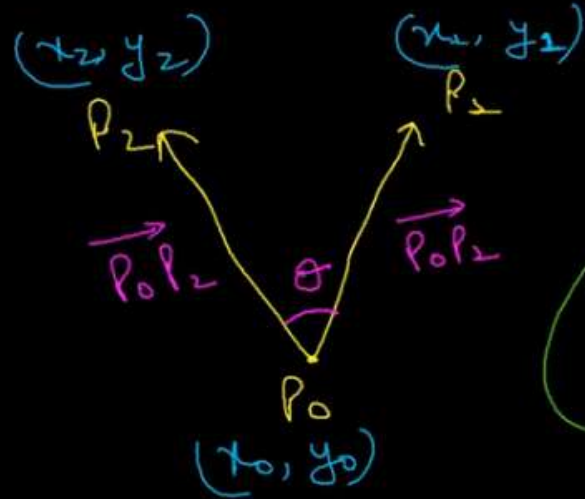
If thumb points <sup>out</sup> ~~above~~ <sup>origin</sup>

<sub>then</sub>  $+VE$

Else if thumb points down  
then  $-VE$

$$\vec{P_1 P_2} = \{ (x_2 - x_1), (y_2 - y_1) \}$$
$$= x_2 y_2 - x_1 y_1$$

# CONCEPT OF CROSS PRODUCT :- (2D) (RH-Rule)



$$\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta$$

$$\vec{P_0P_1} \times \vec{P_0P_2} = \begin{vmatrix} x_1 - x_0 & y_1 - y_0 \\ x_2 - x_0 & y_2 - y_0 \end{vmatrix}$$

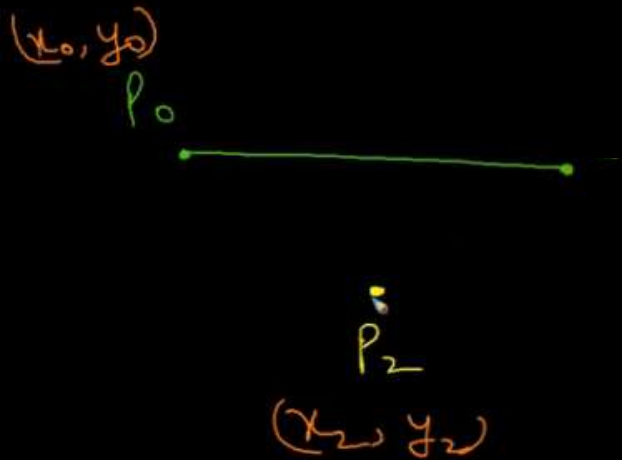
if  $P_0 = \text{Origin}$ ,

then.

$$\vec{P_0P_1} \times \vec{P_0P_2} = \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}$$

In this case, thumb is showing upwards. Hence it is **positive**.

## CONCEPT OF ORIENTATION :-



if  $\vec{P_1P_2} \times \vec{P_1P_0} = \oplus \text{VE}$   
then  $P_2$  is clockwise from  $P_0$   
 $\therefore P_2$  is clockwise from  $P_1$

Else if  $\vec{P_1P_2} \times \vec{P_1P_0} = \ominus \text{VE}$   
then  $P_2$  is Anticlockwise from  $P_0$   
 $\therefore P_2$  is Anticlockwise from  $P_1$

$$\vec{P_1P_2} \times \vec{P_1P_0} = (x_1 - x_2)(y_0 - y_2) - (x_0 - x_2)(y_1 - y_2)$$

what does an orientation mean? let's say you are given a line  $P_0P_1$  and you are given a point  $P_2$  and we want to find the orientation of this point  $P_2$  with respect to this line  $P_0P_1$ , So  
How can we find it ?

$$(x_0, y_0) P_0 \longrightarrow P_1 (x_1, y_1)$$

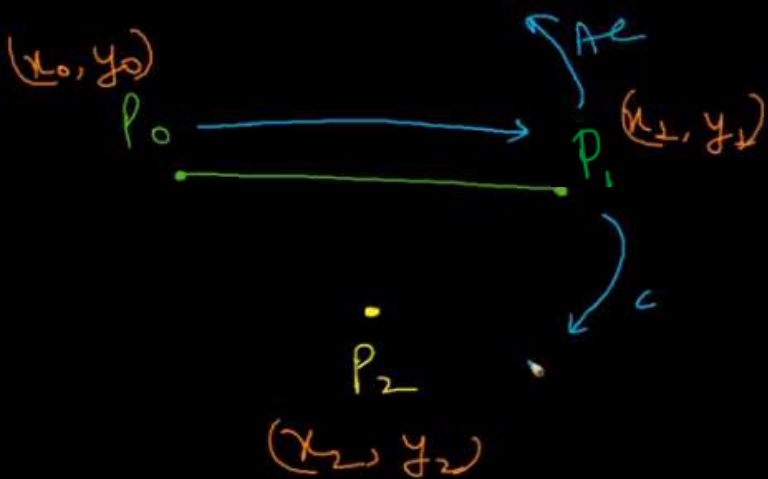
$$P_2 (x_2, y_2) \quad \begin{vmatrix} x_1 - x_2 & x_0 - x_2 \\ y_1 - y_2 & y_0 - y_2 \end{vmatrix}$$

$$\text{Determinant} :- (x_1 - x_2)(y_0 - y_2) - (x_0 - x_2)(y_1 - y_2)$$

$$\vec{P_2 P_1} \times \vec{P_2 P_0} =$$

what does an orientation mean? let's say you are given a line  $P_0 P_1$  and you are given a point  $P_2$  and we want to find the orientation of this point  $P_2$  with respect to this line  $P_0 P_1$ , So  
How can we find it ?

## CONCEPT OF ORIENTATION :-



if  $\vec{P_2P_1} \times \vec{P_2P_0} = \oplus \text{VE}$   
then  $P_1$  is clockwise from  $P_0$   
 $\therefore P_2$  is clockwise from  $P_1$

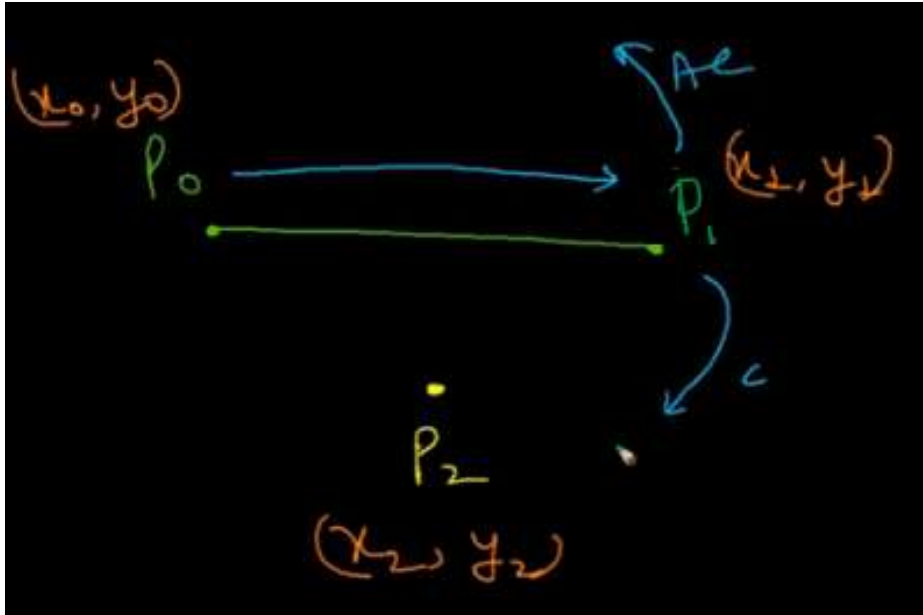
Else if  $\vec{P_2P_1} \times \vec{P_2P_0} = \ominus \text{VE}$   
then  $P_1$  is Anticlockwise from  $P_0$   
 $\therefore P_2$  is Anticlockwise from  $P_1$

$$\vec{P_2P_1} \times \vec{P_2P_0} = (x_1 - x_2)(y_0 - y_2) - (x_0 - x_2)(y_1 - y_2)$$

what does an **orientation** mean? let's say you are given a line  $P_0P_1$  and you are given a point  $P_2$  and we want to find the orientation of this point  $P_2$  with respect to this line  $P_0P_1$ , So

How can we find it ?



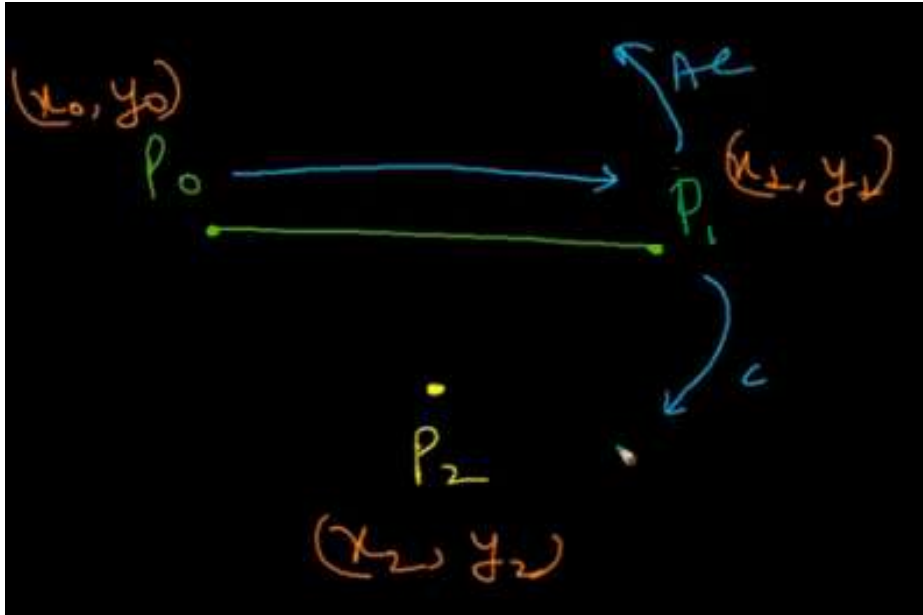


## Orientation of $P_2$ :

let's say you are moving from this point  $P_0P_1$  then if you are at Point  $P_1$  then whether you will move in the **anti-clockwise** direction to reach  $P_2$  or you move in the **clockwise direction** to reach this  $P_2$

in this case you can see that the orientation of  $P_2$  is **clockwise** with respect to vector  $P_0P_1$

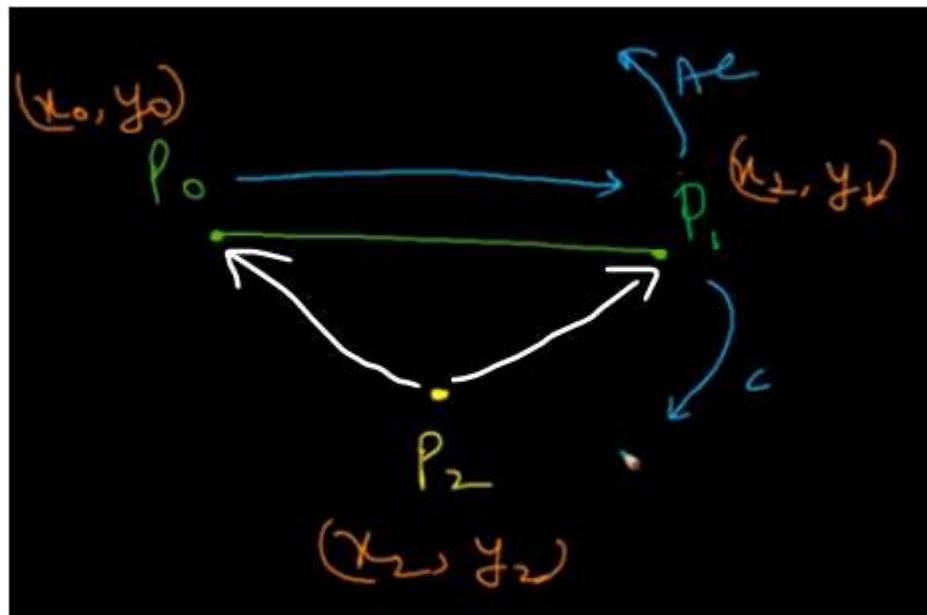
How do we determine this mathematically?



How do we determine this mathematically?

- To do this, draw a vector which is  $P_2P_1$  and another vector  $P_2P_0$
- Take the cross product of these 2 vectors between  $P_2P_1$  and  $P_2P_0$
- Find whether it will be Positive or Negative, then you can **determine the orientation** of this point  $P_2$  with respect to line segment  $P_0P_1$



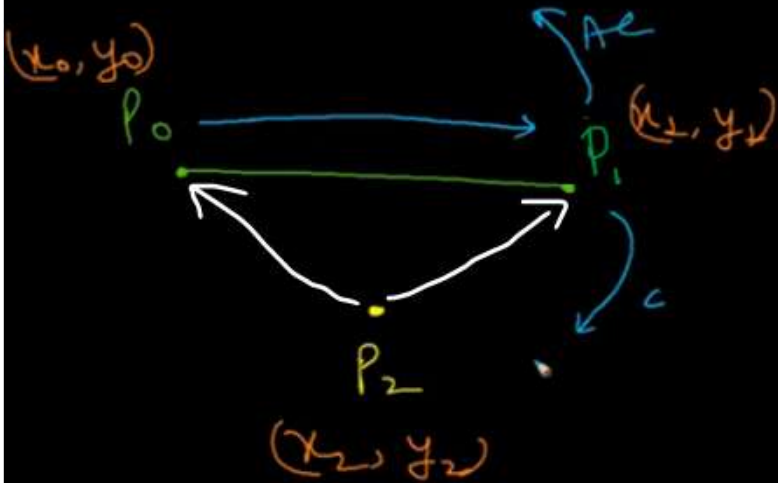


You can also find it out by using the **right hand** rule

If you put your index finger in the direction  $P_2P_1$  and your middle finger in the direction  $P_2P_0$  then the **direction of your thumb** will determine whether the value will come out to be positive or negative.

In this case you might already be knowing that the thumb will be pointing upwards and so the value will be positive and since the value is **positive** therefore from  $P_1$ , We will have to move in the clockwise direction to reach  $P_2$

## CONCEPT OF ORIENTATION :-



if  $\overrightarrow{P_2P_1} \times \overrightarrow{P_2P_0} = \oplus \text{VE}$   
then  $P_1$  is clockwise from  $P_0$   
 $\therefore P_2$  is clockwise from  $P_1$

Else if  $\overrightarrow{P_2P_1} \times \overrightarrow{P_2P_0} = \ominus \text{VE}$   
then  $P_1$  is Anticlockwise from  $P_0$   
 $\therefore P_2$  is Anticlockwise from  $P_1$

Cross product = Find determinant

## Cross Product : Determinant Formula

FOR LINE SEGMENTS  $\overrightarrow{P_0 P_1}$  AND  $\overrightarrow{P_0 P_2}$

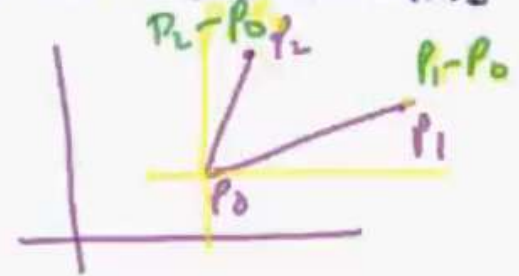
TO FIND IF  $\overrightarrow{P_0 P_1}$  IS CLOCK WISE / COUNTER CLOCKWISE

SHIFT  $P_0$  TO BE ORIGIN

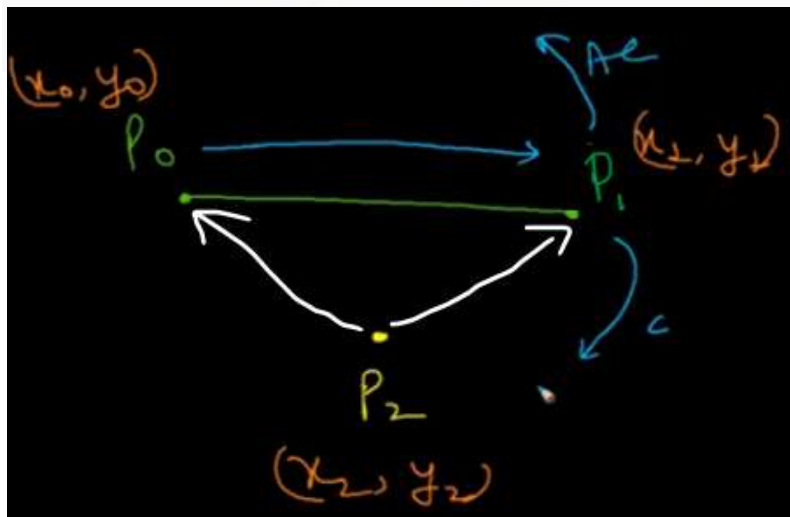
$$\Rightarrow \vec{P_1'} = (P_1 - P_0) \quad \vec{P_2'} = (P_2 - P_0)$$

$$\vec{P_1'} \times \vec{P_2'} = (P_1 - P_0) \times (P_2 - P_0)$$

$$= \begin{vmatrix} x_1 - x_0 & y_1 - y_0 \\ x_2 - x_0 & y_2 - y_0 \end{vmatrix} = (x_1 - x_0)(y_2 - y_0) - (x_2 - x_0)(y_1 - y_0)$$



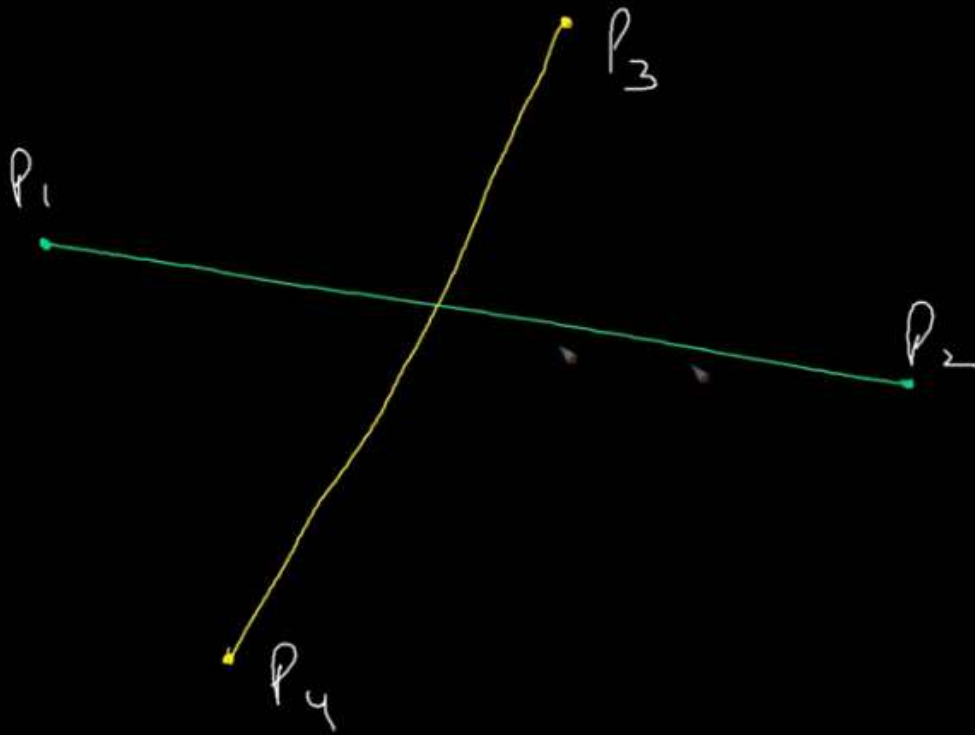
IF  $\vec{P_1'} \times \vec{P_2'} > 0$   $\overrightarrow{P_0 P_1}$  IS CLOCKWISE FROM  $\overrightarrow{P_0 P_2}$  ELSE COUNTER CLOCK WISE



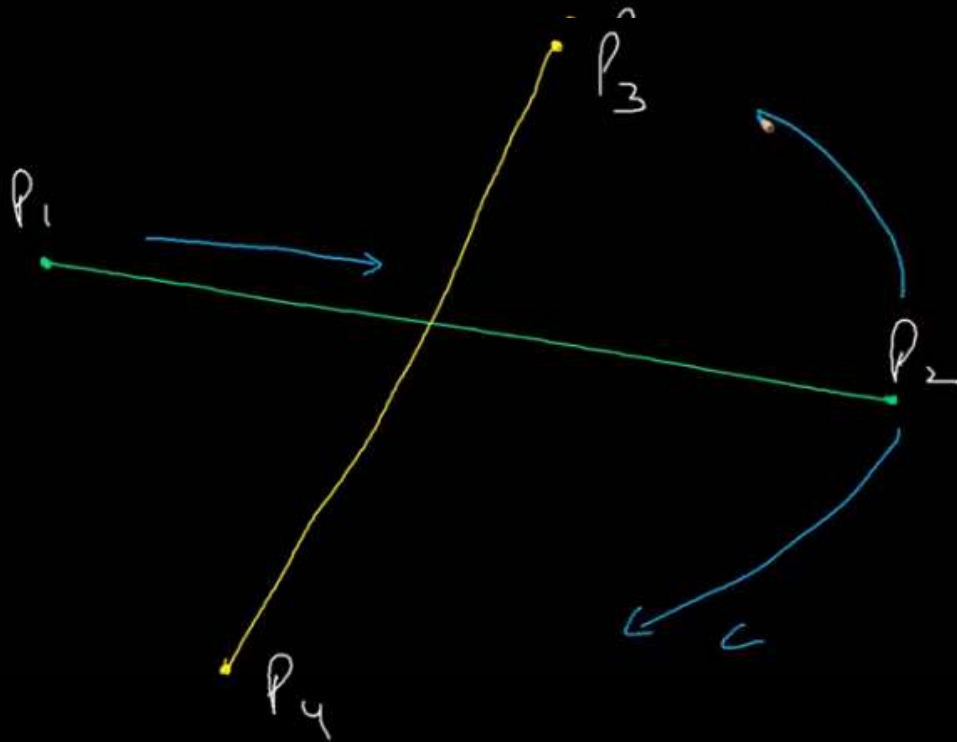
X



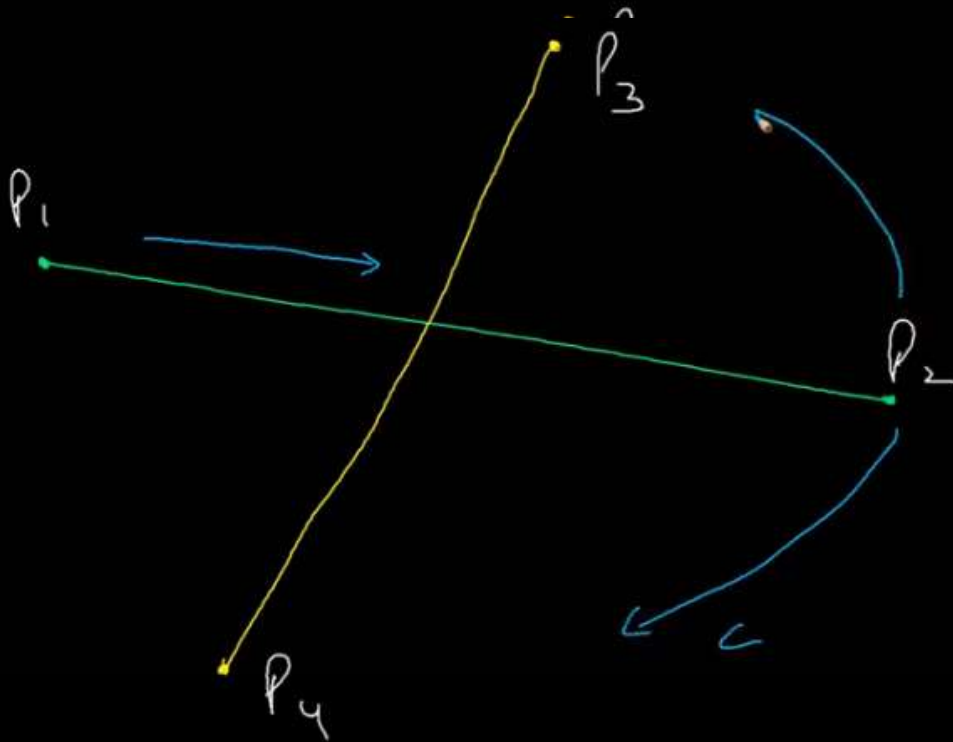
Let us solve the problem using these concepts



1. Pick one line :  $P_1P_2$
2. Need to find the orientation of 2 points of another line.  
i.e. Orientation of  $P_4$  with respect to  $P_1P_2$   
Orientation of  $P_3$  with respect to  $P_1P_2$



3. when we draw a vector  $P_1P_2$  then from  $P_2$  you are required to move in the **clockwise** direction to reach  $P_4$  and
4. you will have to move in the **anti-clockwise** direction to reach  $P_3$



**i) Pick a line  $P_1P_2$ .**

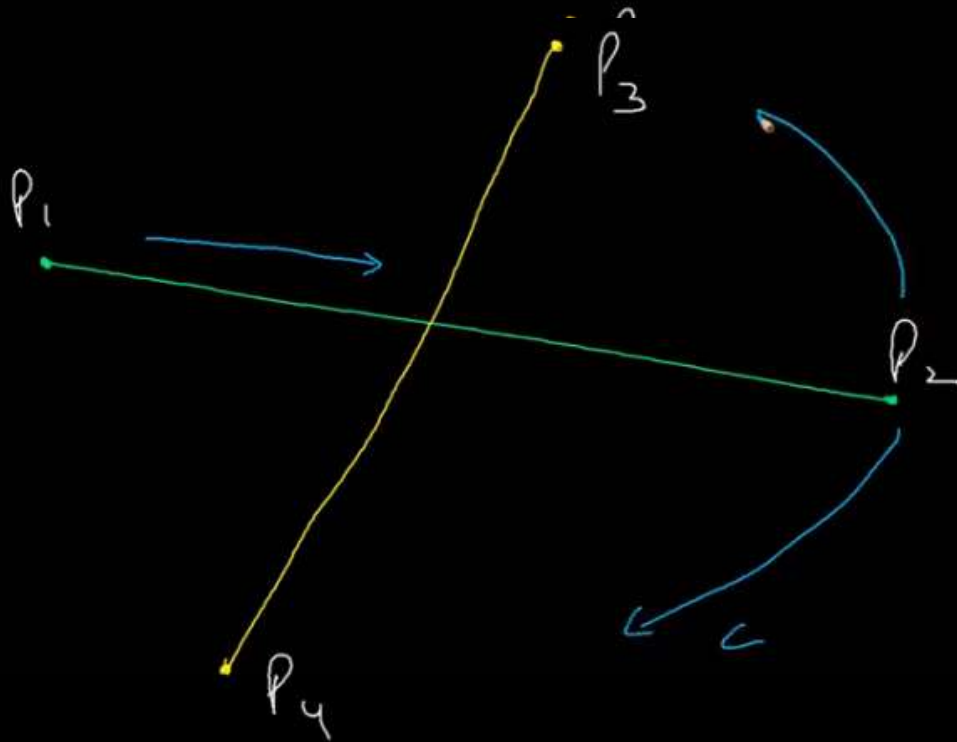
Orientation of  $P_4$  with respect to  $P_1P_2$

Orientation of  $P_4$ : From  $P_2$ , Move Clockwise to reach  $P_4$

Orientation of  $P_3$  with respect to  $P_1P_2$

Orientation of  $P_3$ : From  $P_2$ , Move Anticlockwise to reach  $P_3$

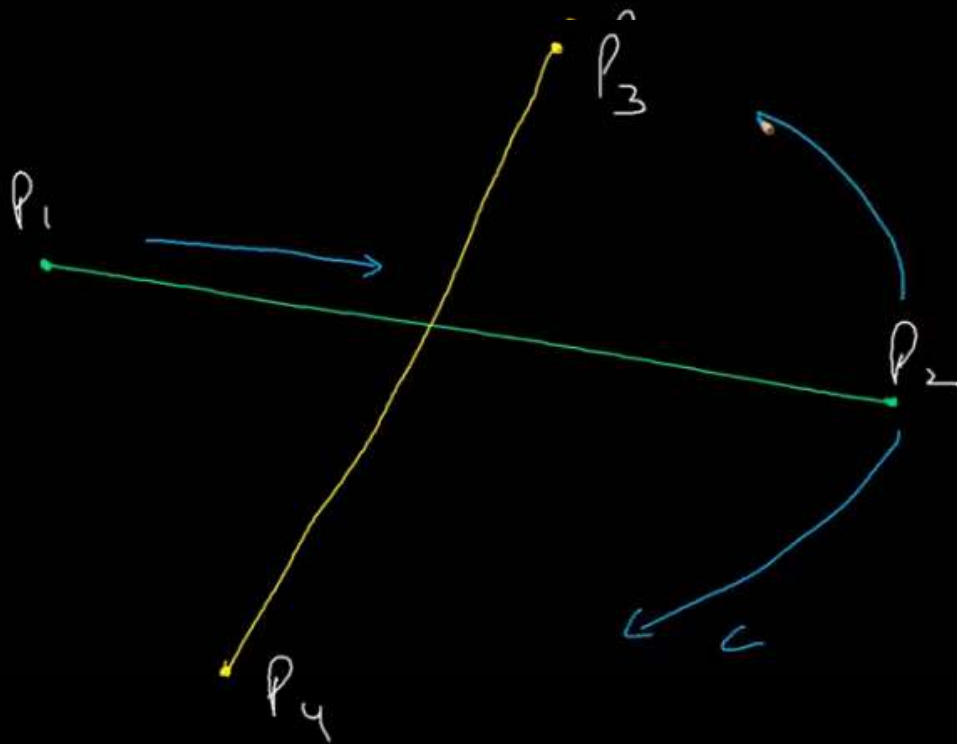




**Inference:** this is very important these two points should be in **opposite direction**

if one is in clockwise direction then the another one should always be in anti-clockwise direction

once you determine that if **both are in anti-clockwise** **or** both are in **clockwise** then those two line segments **can never intersect**



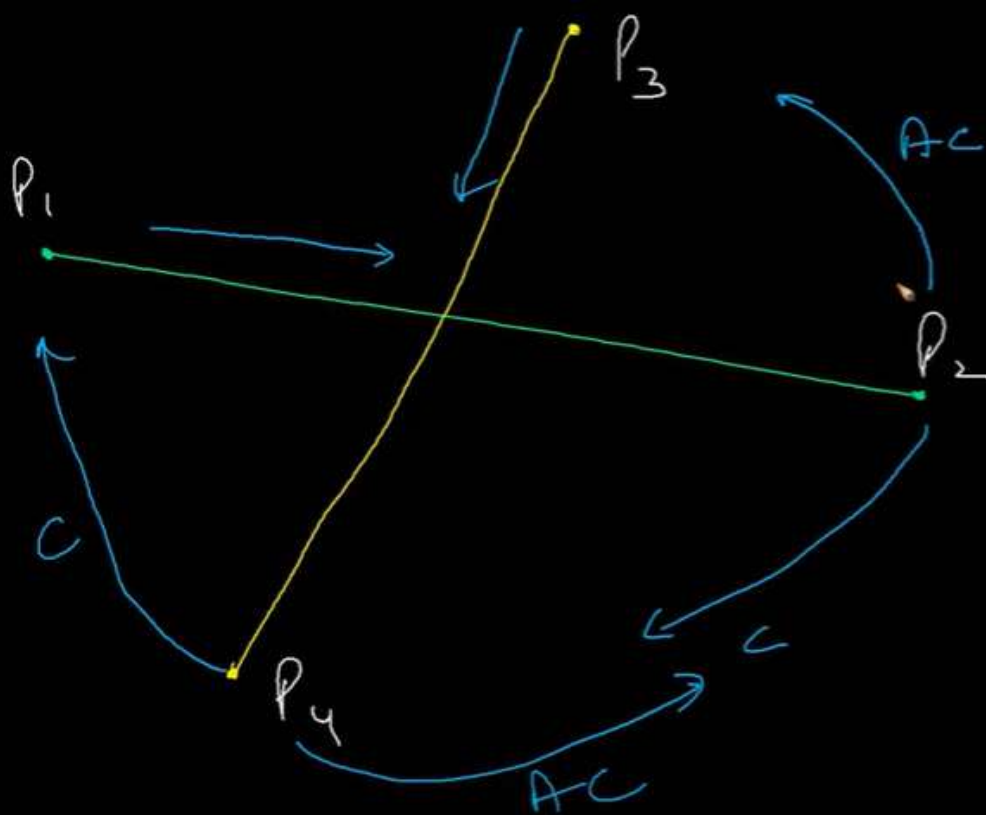
ii) Pick another line  $P_3P_4$

Find orientation of  $P_1$  w.r.t  $P_3P_4$ .

Orientation of  $P_1$ : From  $P_4$ , Move **Clockwise** to reach  $P_1$

Find orientation of  $P_2$  w.r.t  $P_3P_4$ .

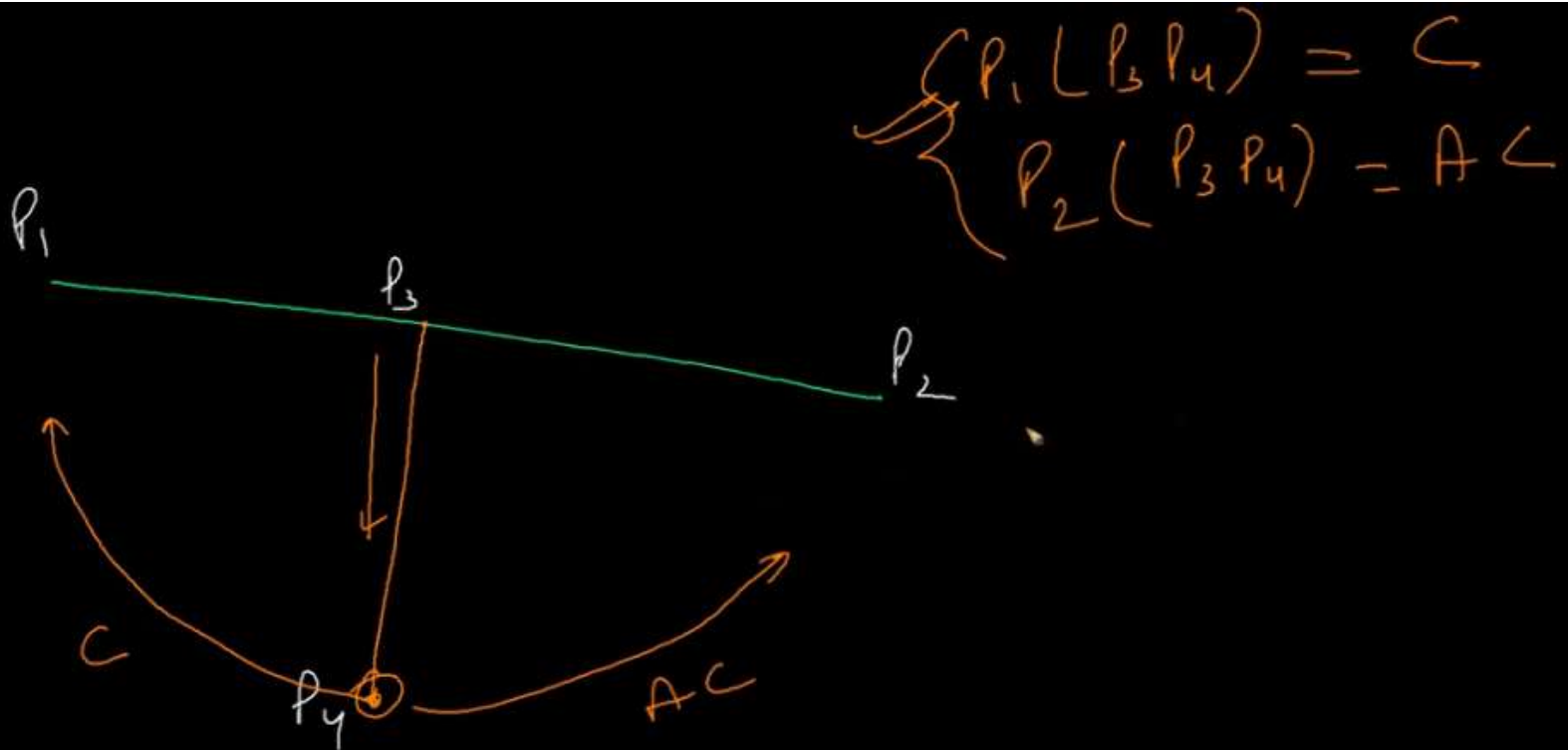
Orientation of  $P_2$ : From  $P_4$ , Move **AntiClockwise** to reach  $P_2$



$$\begin{aligned} & \textcircled{\underline{P_1 P_2}} \\ & \begin{cases} P_4(P_1 P_2) = C \\ P_3(P_1 P_2) = AC \end{cases} \\ & \begin{cases} P_1(P_3 P_4) = C \\ P_2(P_3 P_4) = AC \end{cases} \end{aligned}$$

- Once we confirm that **these rules hold** for both these lines,
- And both the points orient in different direction, then we can **confirm that the line segments intersect**.

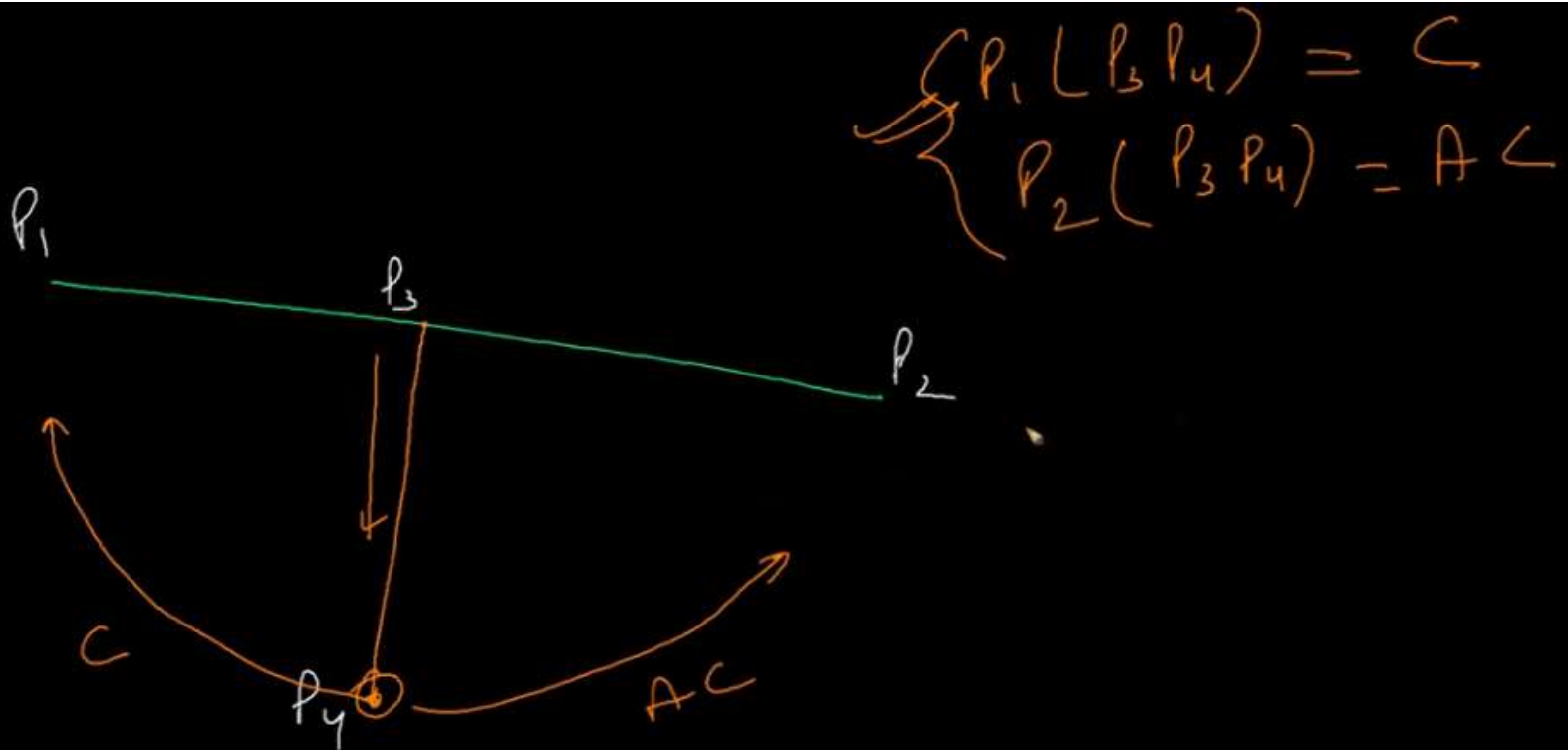
# Another Example



### i) Pick line $P_3P_4$

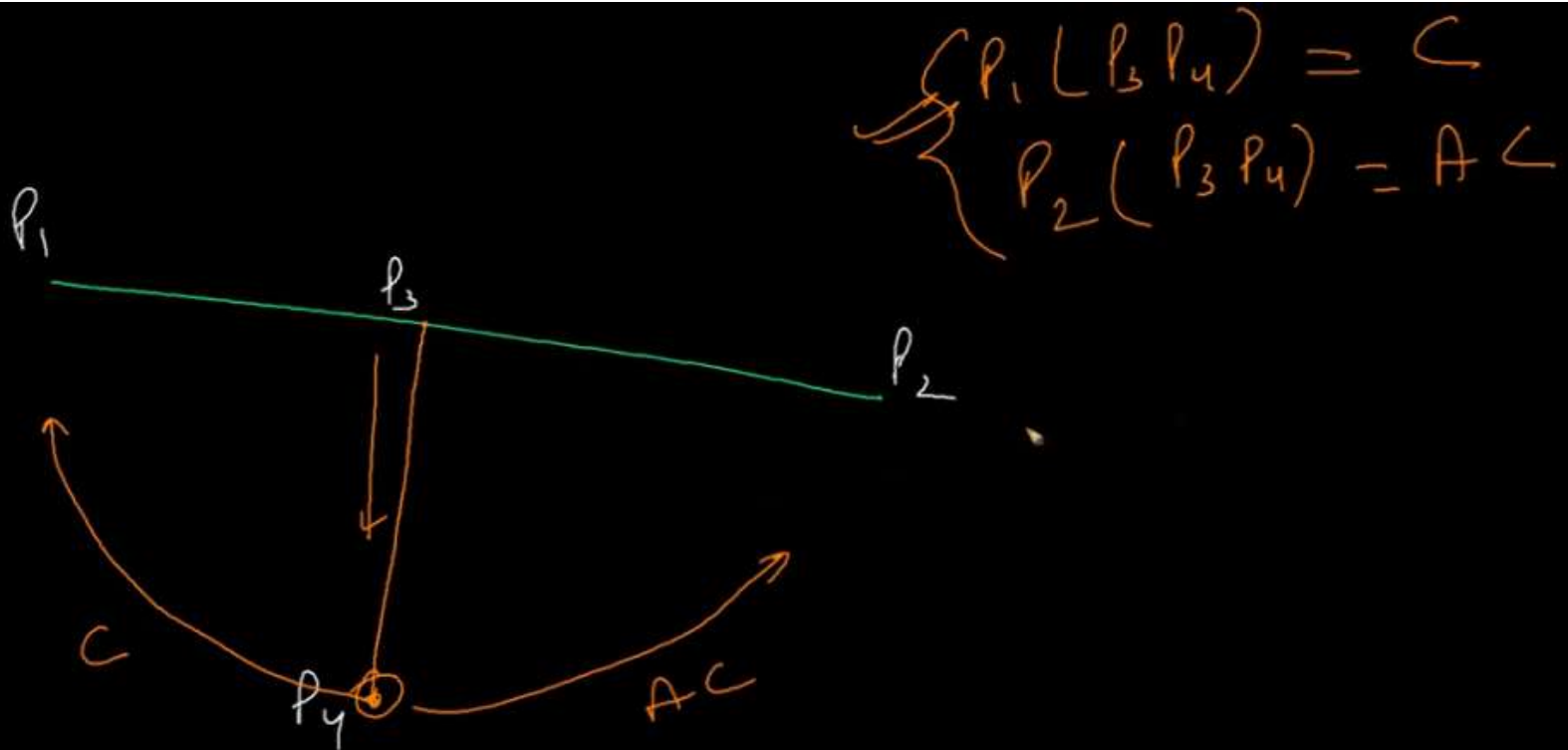
Find orientation of  $P_1$  w.r.t  $P_3P_4$ .

Orientation of  $P_1$ : From  $P_4$ , Move **Clockwise** to reach  $P_1$

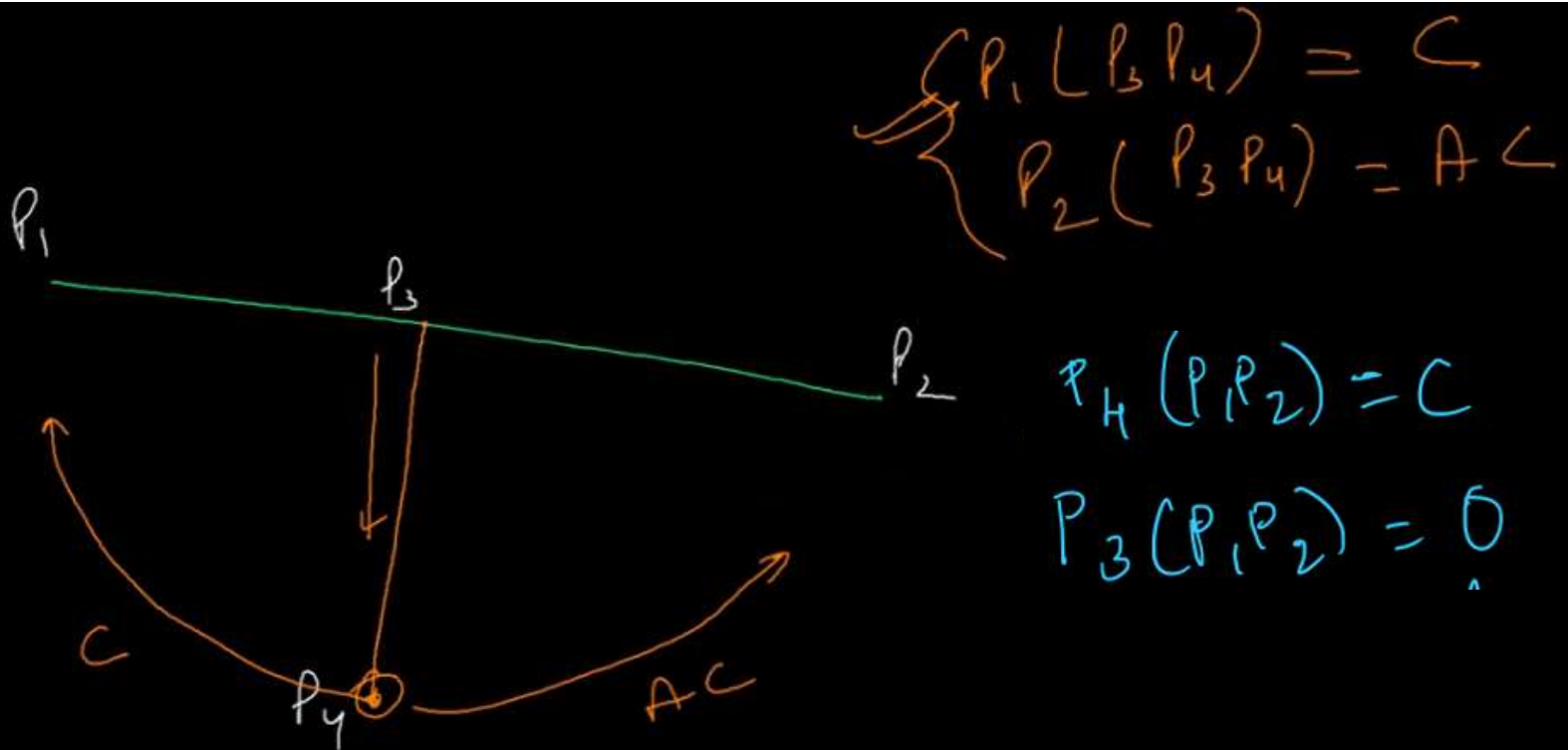


Find orientation of  $P_2$  w.r.t  $P_3P_4$ .

Orientation of  $P_2$ : From  $P_4$ , Move **AntiClockwise** to reach  $P_2$



Hence, when we pick the line  $P_3P_4$ , the orientation of the 2 points  $P_1$  and  $P_2$  are in **opposite direction** from the given point  $P_4$ .  
 So, this **holds our condition**



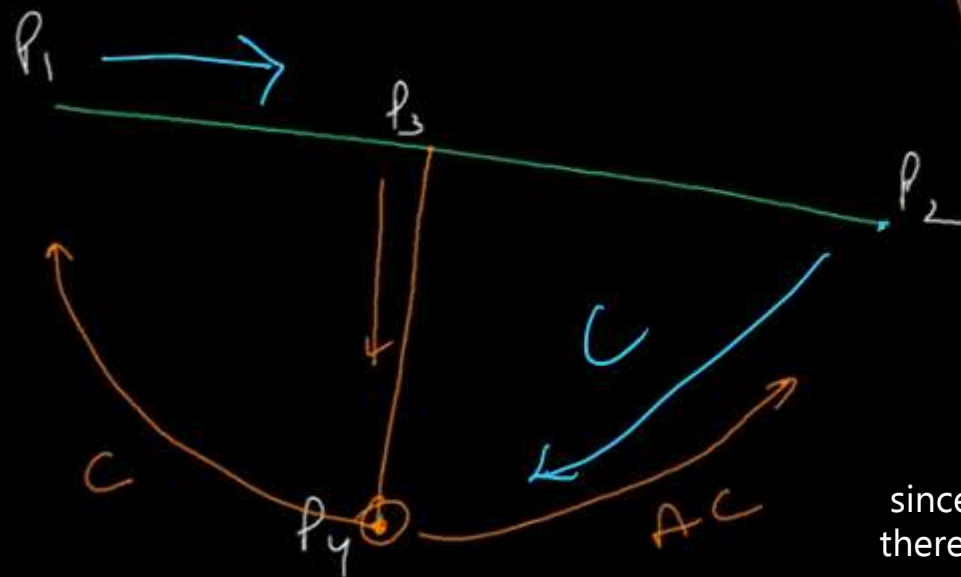
ii) Pick another line  $P_1 P_2$ .

Find the orientation of  $P_3$  and  $P_4$ .

Orientation of  $P_4$ : From  $P_2$ , Move Clockwise

Orientation of  $P_3$ : From  $P_1$ , ?





$$\begin{cases} P_1(P_3, P_4) = C \\ P_2(P_3, P_4) = A < \end{cases}$$

$$P_H(P_1, P_2) = C$$

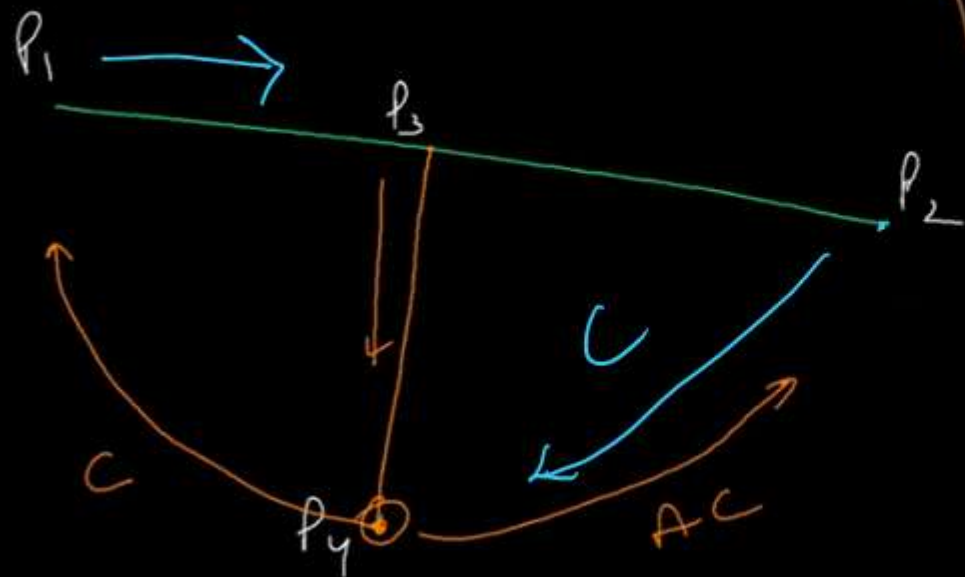
$$P_3(P_1, P_2) = 0$$

since it is lying on the same line segment  $P_1P_2$  therefore the **cross-product** will come out to be 0, since  $\theta = 180$  degrees,  $\sin \theta = 0$  while we are using this in cross product.

**whenever** this comes out to be 0, i.e. **orientation 0** then we **always assume** that it is **always opposite** to our **found orientation C** (so in this case we determined that  $P_4$  had an orientation of clockwise) and  $p_3$  has orientation of **0** and so we will take 0 as anti-clockwise which is **opposite** to this C

Orientation of  $P_4$ : From  $P_2$ , Move Clockwise

Orientation of  $P_3$ : From  $P_1$ , **0**



$$\begin{cases} P_1(P_3P_4) = C \\ P_2(P_3P_4) = A < \end{cases}$$

$$P_H(P_1P_2) = C$$

$$P_3(P_1P_2) = 0$$



Anticlockwise

**whenever** this comes out to be 0, i.e. **orientation 0** then we **always assume** that it is **always opposite** to our **found orientation C** (so in this case we determined that  $P_4$  had an orientation of clockwise) and  $p_3$  has orientation of **0** and so we will take 0 as **anti-clockwise** which is **opposite** to this C.

Hence, this also holds our condition. Hence, **2 line segments intersect**.

# Another Example - Not intersect

**i) Pick a line  $P_1P_2$ .**

Orientation of  $P_4$  with respect to  $P_1P_2$

Orientation of  $P_4$ : From  $P_2$ , Move Clockwise

Orientation of  $P_3$  with respect to  $P_1P_2$

Orientation of  $P_3$ : From  $P_2$ , Move Anticlockwise

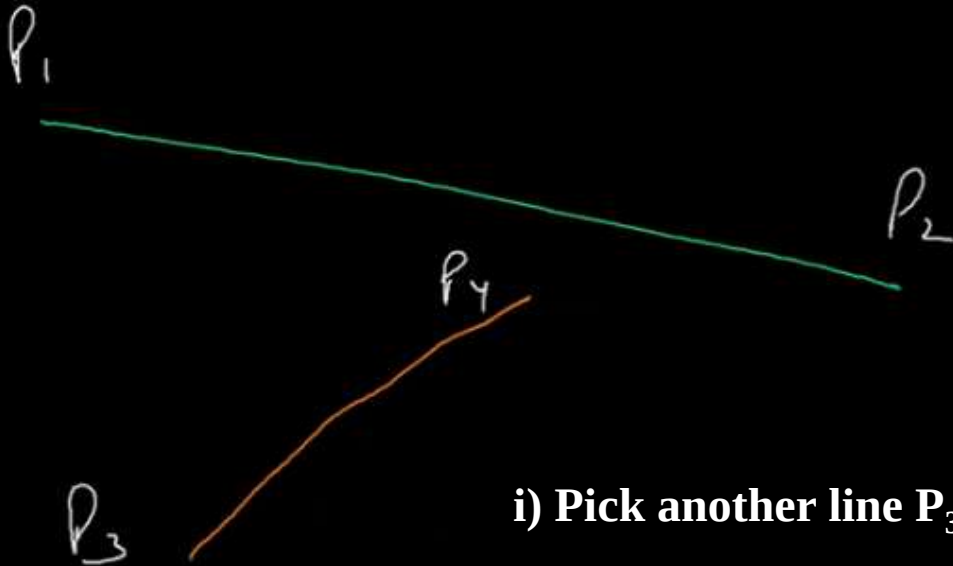
**ii) Pick another line  $P_3P_4$**

Find orientation of  $P_1$  w.r.t  $P_3P_4$ .

Orientation of  $P_1$ : From  $P_4$ , Move **Clockwise** to reach  $P_1$

Find orientation of  $P_2$  w.r.t  $P_3P_4$ .

Orientation of  $P_2$ : From  $P_4$ , Move **AntiClockwise** to reach  $P_2$



i) Pick another line  $P_3P_4$

Find orientation of  $P_1$  w.r.t  $P_3P_4$ .

Orientation of  $P_1$ : From  $P_4$ , Move **Clockwise** to reach  $P_1$

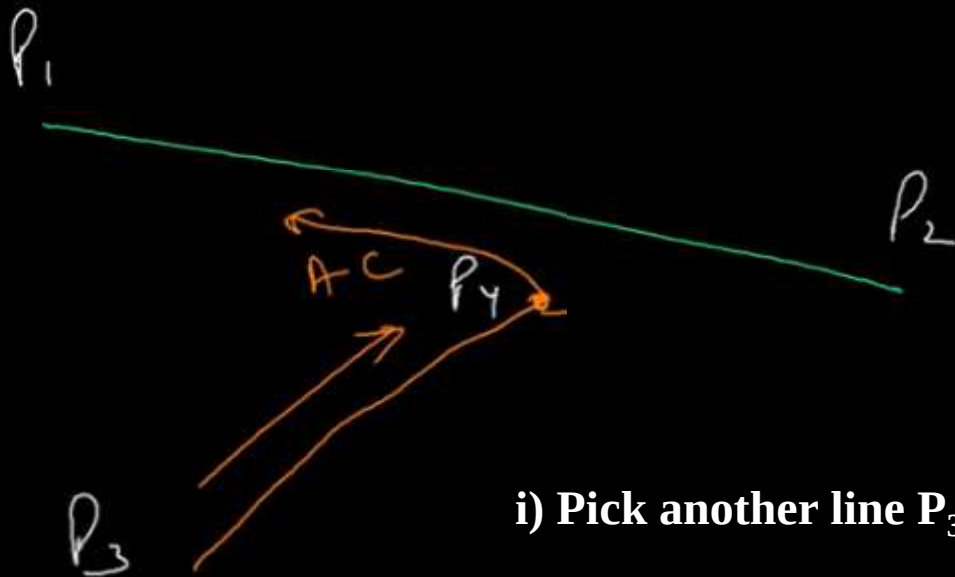
Find orientation of  $P_2$  w.r.t  $P_3P_4$ .

Orientation of  $P_2$ : From  $P_4$ , Move **AntiClockwise** to reach  $P_2$

**1. Pick line  $P_3P_4$**

Find orientation of  $P_1$  w.r.t line segment  $P_3P_4$ .

Orientation of  $P_1$ : From  $P_4$ , Move **?**



i) Pick another line  $P_3P_4$

Find orientation of  $P_1$  w.r.t  $P_3P_4$ .

Orientation of  $P_1$ : From  $P_4$ , Move **Clockwise** to reach  $P_1$

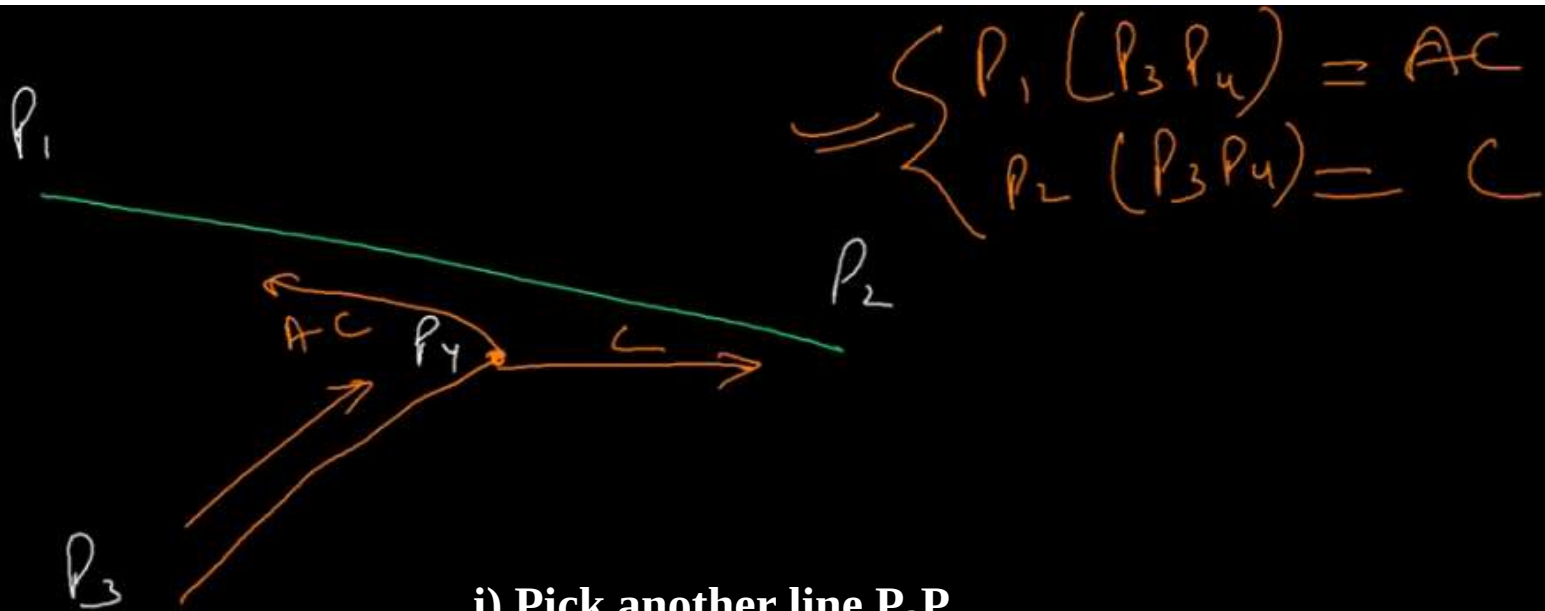
Find orientation of  $P_2$  w.r.t  $P_3P_4$ .

Orientation of  $P_2$ : From  $P_4$ , Move **AntiClockwise** to reach  $P_2$

1. Pick line  $P_3P_4$

Find orientation of  $P_1$  w.r.t line segment  $P_3P_4$ .

Orientation of  $P_1$ : From  $P_4$ , Move **AntiClockwise** to reach  $P_1$



i) Pick another line  $P_3P_4$

Find orientation of  $P_1$  w.r.t  $P_3P_4$ .

Orientation of  $P_1$ : From  $P_4$ , Move **Clockwise** to reach  $P_1$

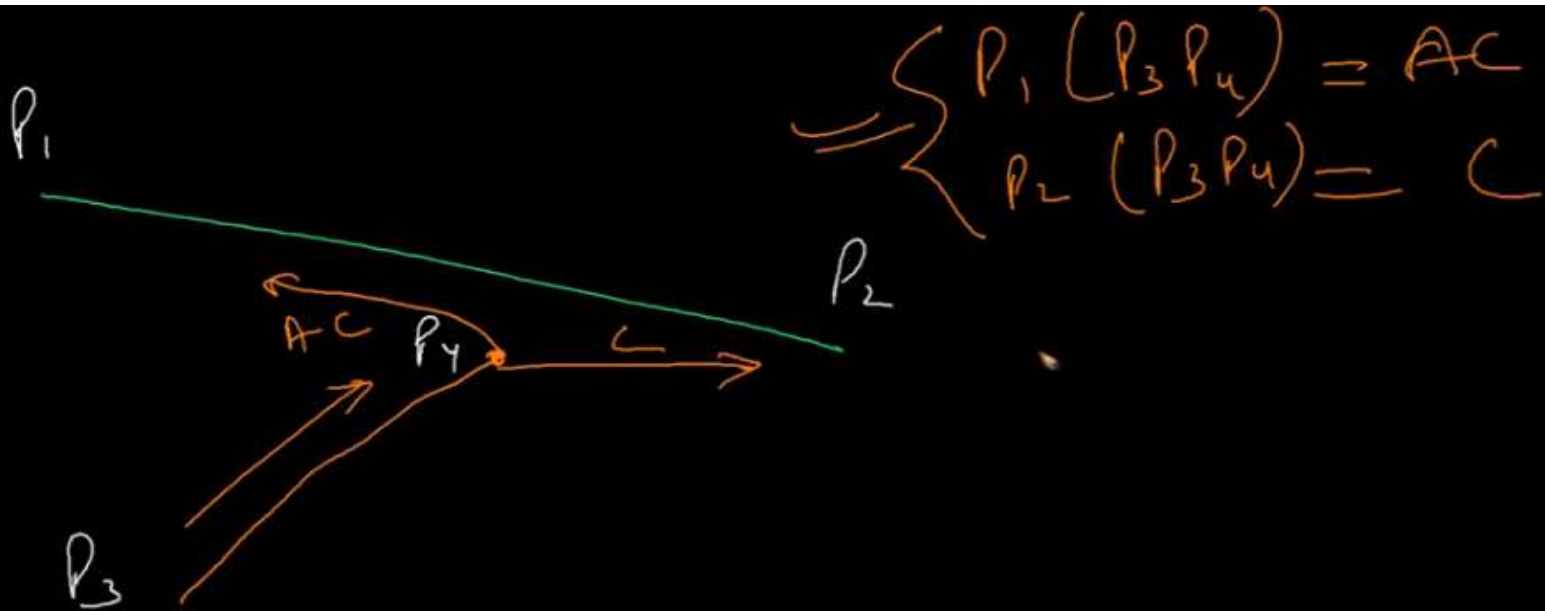
Find orientation of  $P_2$  w.r.t  $P_3P_4$ .

Orientation of  $P_2$ : From  $P_4$ , Move **AntiClockwise** to reach  $P_2$

Find orientation of  $P_2$  w.r.t  $P_3P_4$ .

Orientation of  $P_2$ : From  $P_4$ , Move **Clockwise** to reach  $P_2$

Hence, this holds our **condition**.



ii) Pick a line  $P_1P_2$ .

Orientation of  $P_4$  with respect to  $P_1P_2$

Orientation of  $P_4$ : From  $P_2$ , Move Clockwise

Orientation of  $P_3$  with respect to  $P_1P_2$

Orientation of  $P_3$ : From  $P_2$ , Move Anticlockwise

ii) Pick **another** line  $P_1P_2$ .

**Find** the orientation of  $P_3$  and  $P_4$ .

Orientation of  $P_4$ : From  $P_2$ , Move Clockwise

Orientation of  $P_3$ : From  $P_2$ , Move **Clockwise**

$P_1$

$$\begin{cases} P_1(P_3 P_4) = AC \\ P_2(P_3 P_4) = C \end{cases}$$

$P_2$

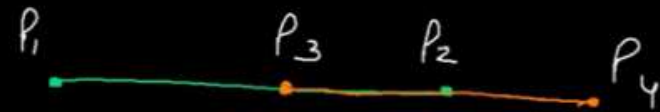
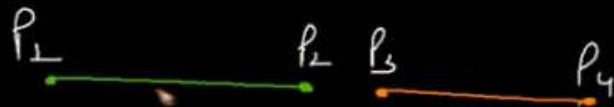


$P_3$



## SPECIAL CASES :- (COLLINEAR)

$$\begin{aligned} P_1 &= (x_1, y_1) & P_3 &= (x_3, y_3) \\ P_2 &= (x_2, y_2) & P_4 &= (x_4, y_4) \end{aligned}$$



Orientation = 0  $\therefore \theta = 180^\circ$   $\left( \vec{a} \times \vec{b} = ab \sin \theta = 0 \right)$

Now if Either  $P_3$  lies in range of  $(P_1 P_2)$  OR  $P_4$  " " " " then OVERLAPPING Else No overlap  $\therefore$  No intersection.



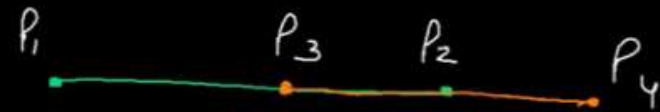
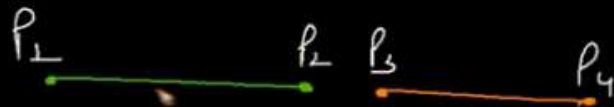
$$\text{if } ((x_3 \geq x_1 \wedge x_3 \leq x_2) \wedge (y_3 \geq y_1 \wedge y_3 \leq y_2))$$

Case 1) Collinear not intersecting (No overlap)

Case 2) Collinear also overlapping (Intersecting)

## SPECIAL CASES :- (COLLINEAR)

$$\begin{aligned} P_1 &= (x_1, y_1) & P_3 &= (x_3, y_3) \\ P_2 &= (x_2, y_2) & P_4 &= (x_4, y_4) \end{aligned}$$



Orientation = 0  $\therefore \theta = 180^\circ$   $\left( \vec{a} \times \vec{b} = ab \sin \theta = 0 \right)$

Now if Either  $P_3$  lies in range of  $(P_1 P_2)$  OR  $P_4$  " " " " " " then OVERLAPPING Else No overlap  $\therefore$  No intersection.



$$\text{if } ( (x_3 \geq x_1 \wedge x_3 \leq x_2) \wedge (y_3 \geq y_1 \wedge y_3 \leq y_2) )$$

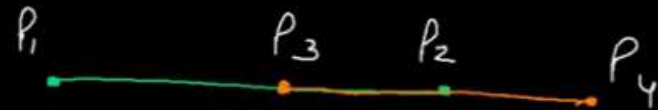
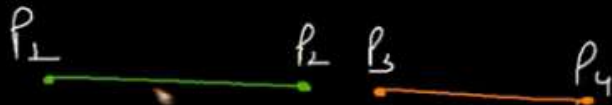
Orientation of  $P_3$  w.r.t  $P_1 P_2 = 0$

Orientation of  $P_4$  w.r.t  $P_1 P_2 = 0$  since  $\theta = 180$ , hence crossProd =

0

## SPECIAL CASES :- (COLLINEAR)

$$\begin{array}{ll} P_1 = (x_1, y_1) & P_3 = (x_3, y_3) \\ P_2 = (x_2, y_2) & P_4 = (x_4, y_4) \end{array}$$



Orientation = 0  $\therefore \theta = 180^\circ$   $\left( \vec{a} \times \vec{b} = ab \sin \theta \right)$   
 $= 0$

Now, if either  $P_3$  lies in range of  $(P_1 P_2)$   
OR  $P_4$  " " " "  
then OVERLAPPING Else No overlap  
 $\therefore$  No intersection.  
if  $( (x_3 \geq x_1 \wedge x_3 \leq x_2) \wedge (y_3 \geq y_1 \wedge y_3 \leq y_2) )$



1. you find orientation equals to 0 for both the points, let's say for both points  $p_3$  and  $p_4$ , then you can **confirm** that both these line segments are **collinear**.

so our first requirement is confirmed which is for collinearity

2. **Next step** is : have 2 options: whether they overlap or not

# Algorithm

# Algorithm

```
struct Point
{
    int x;
    int y;
};
```

```
// Given three collinear points p, q, r, the function checks if
// point q lies on line segment 'pr'
```

```
bool onSegment(Point p, Point q, Point r)
{
    if (q.x <= max(p.x, r.x) && q.x >= min(p.x, r.x) &&
        q.y <= max(p.y, r.y) && q.y >= min(p.y, r.y))
        return true;

    return false;
}
```

# Algorithm (cont...)

```
// To find orientation of ordered triplet (p, q, r).  
// The function returns following values  
// 0 --> p, q and r are collinear  
// 1 --> Clockwise  
// 2 --> Counterclockwise
```

```
int orientation(Point p, Point q, Point r)  
{  
    int val = (q.y - p.y) * (r.x - q.x) -  
              (q.x - p.x) * (r.y - q.y);  
  
    if (val == 0) return 0; // collinear  
  
    return (val > 0)? 1: 2; // clock or counterclock wise  
}
```

## 1. Vector Representation:

- $\vec{pq} = (q_x - p_x, q_y - p_y)$
- $\vec{qr} = (r_x - q_x, r_y - q_y)$

$$\vec{pq} \times \vec{qr} = \begin{bmatrix} (q_x - p_x) & (r_x - q_x) \\ (q_y - p_y) & (r_y - q_y) \end{bmatrix}$$

$$\vec{pq} \times \vec{qr} = (q_x - p_x) \cdot (r_y - q_y) - (q_y - p_y) \cdot (r_x - q_x)$$



# Algorithm (cont...)

## 1. Vector Representation:

- $\vec{pq} = (q_x - p_x, q_y - p_y)$
- $\vec{qr} = (r_x - q_x, r_y - q_y)$

Find Orientation of r w.r.t pq

## 2. Cross Product Formula:

The z-component of the cross product  $\vec{pq} \times \vec{qr}$  is:

$$\vec{pq} \times \vec{qr} = \begin{bmatrix} (q_x - p_x) & (r_x - q_x) \\ (q_y - p_y) & (r_y - q_y) \end{bmatrix}$$

$$\vec{pq} \times \vec{qr} = (q_x - p_x) \cdot (r_y - q_y) - (q_y - p_y) \cdot (r_x - q_x)$$

But in the code, it's written as  $(q_y - p_y) \cdot (r_x - q_x) - (q_x - p_x) \cdot (r_y - q_y)$ . This is the same expression with terms rearranged due to the antisymmetric property of the cross product. To see this, expand both:

- Original:  $(q_x - p_x)(r_y - q_y) - (q_y - p_y)(r_x - q_x)$
- Code version:  $(q_y - p_y)(r_x - q_x) - (q_x - p_x)(r_y - q_y)$

These are negatives of each other because swapping the order of the vectors in a cross product negates the result. The sign convention depends on the order of points, but the logic remains consistent with the orientation check.

# Algorithm (cont...)

## 3. Interpretation of the Result:

- If  $\text{val} = 0$ : The points are **collinear** (the vectors  $\vec{qp}$  and  $\vec{qr}$  are parallel, so  $r$  lies on the line through  $p$  and  $q$ ).
- If  $\text{val} > 0$ : The orientation is **clockwise** ( $r$  is to the right of the vector  $\vec{pq}$  when moving from  $p$  to  $q$ ).
- If  $\text{val} < 0$ : The orientation is **counterclockwise** ( $r$  is to the left of the vector  $\vec{pq}$  when moving from  $p$  to  $q$ ).

## Geometric Intuition

The cross product's z-component determines the signed area of the parallelogram formed by  $\vec{qp}$  and  $\vec{qr}$ . This area's sign indicates the relative orientation:

- A positive value means  $r$  is on the counterclockwise side of  $\vec{pq}$ .
- A negative value means  $r$  is on the clockwise side.
- Zero means  $r$  is on the same line as  $p$  and  $q$ .



# Algorithm (cont...)

```
// The main function that returns true if line segment 'p1q1'
// and 'p2q2' intersect.
bool doIntersect(Point p1, Point q1, Point p2, Point q2)
{
    // Find the four orientations needed for general and
    // special cases
    int o1 = orientation(p1, q1, p2);
    int o2 = orientation(p1, q1, q2);
    int o3 = orientation(p2, q2, p1);
    int o4 = orientation(p2, q2, q1);

    // General case
    if (o1 != o2 && o3 != o4)
        return true;
```

# Algorithm (cont...)

```
// Special Cases
```

```
// p1, q1 and p2 are collinear and p2 lies on segment p1q1
```

```
if (o1 == 0 && onSegment(p1, p2, q1)) return true;
```

```
// p1, q1 and q2 are collinear and q2 lies on segment p1q1
```

```
if (o2 == 0 && onSegment(p1, q2, q1)) return true;
```

```
// p2, q2 and p1 are collinear and p1 lies on segment p2q2
```

```
if (o3 == 0 && onSegment(p2, p1, q2)) return true;
```

```
    // p2, q2 and q1 are collinear and q1 lies on segment p2q2
```

```
if (o4 == 0 && onSegment(p2, q1, q2)) return true;
```

```
return false; // Doesn't fall in any of the above cases
```

```
}
```

# Algorithm (cont...)

## Test Data :

```
struct Point p1 = {1, 1}, q1 = {10, 1};
```

```
struct Point p2 = {1, 2}, q2 = {10, 2};
```

```
doIntersect(p1, q1, p2, q2)? cout << "Yes\n": cout << "No\n";
```

```
p1 = {10, 0}, q1 = {0, 10};
```

```
p2 = {0, 0}, q2 = {10, 10};
```

```
doIntersect(p1, q1, p2, q2)? cout << "Yes\n": cout << "No\n";
```

```
p1 = {-5, -5}, q1 = {0, 0};
```

```
p2 = {1, 1}, q2 = {10, 10};
```

```
doIntersect(p1, q1, p2, q2)? cout << "Yes\n": cout << "No\n";
```

# Algorithm (cont...)

Output:

## Test Data :

```
struct Point p1 = {1, 1}, q1 = {10, 1};  
struct Point p2 = {1, 2}, q2 = {10, 2};
```

```
doIntersect(p1, q1, p2, q2)? cout << "Yes\n": cout << "No\n";
```

```
p1 = {10, 0}, q1 = {0, 10};  
p2 = {0, 0}, q2 = {10, 10};  
doIntersect(p1, q1, p2, q2)? cout << "Yes\n": cout << "No\n";
```

```
p1 = {-5, -5}, q1 = {0, 0};  
p2 = {1, 1}, q2 = {10, 10};  
doIntersect(p1, q1, p2, q2)? cout << "Yes\n": cout << "No\n";
```

*Time Complexity:  $O(1)$*

*Space Complexity:  $O(1)$*

No

Yes

No

# How to check if two given line segments intersect?

Given two line segments  $(p1, q1)$  and  $(p2, q2)$ , find if the given line segments intersect with each other.

Before we discuss the solution, let us define notion of **orientation**.

Orientation of an ordered triplet of points in the plane can be

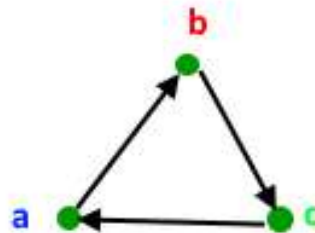
–**counterclockwise**

–**clockwise**

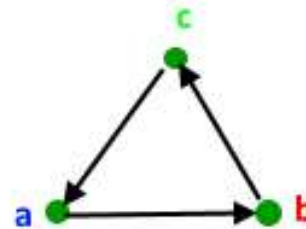
–**collinear**

The following diagram shows different possible orientations of

$(a, b, c)$



**Clockwise**



**Counterclockwise**



**Collinear**

# How is Orientation useful here?

Two segments  $(p1, q1)$  and  $(p2, q2)$  intersect if and only if one of the following two conditions is verified

## 1. *General Case:*

- $(p1, q1, p2)$  and  $(p1, q1, q2)$  have different orientations and
- $(p2, q2, p1)$  and  $(p2, q2, q1)$  have different orientations.

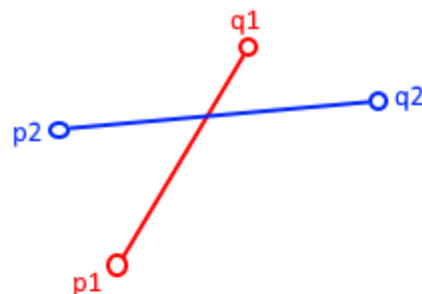
Examples:

# How is Orientation useful here?

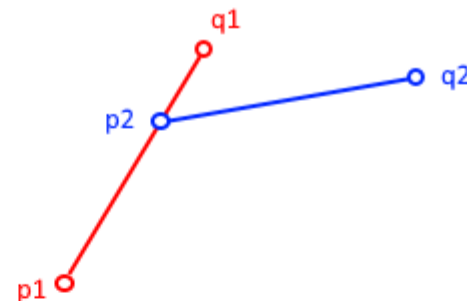
## 1. General Case:

- $(p1, q1, p2)$  and  $(p1, q1, q2)$  have different orientations and
- $(p2, q2, p1)$  and  $(p2, q2, q1)$  have different orientations.

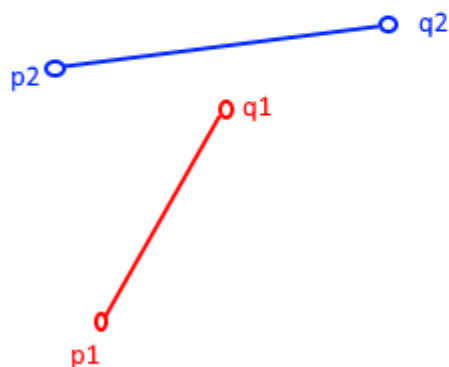
Examples:



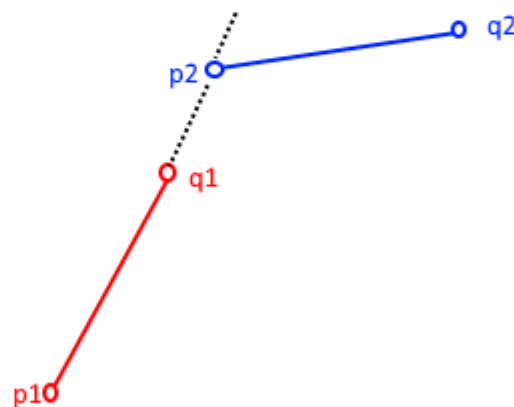
**Example :** Orientations of  $(p1, q1, p2)$  and  $(p1, q1, q2)$  are different. Orientations of  $(p2, q2, p1)$  and  $(p2, q2, q1)$  also different.



**Example:** Orientations of  $(p1, q1, p2)$  and  $(p1, q1, q2)$  are different. Orientations of  $(p2, q2, p1)$  and  $(p2, q2, q1)$  also different



**Example:** Orientations of  $(p1, q1, p2)$  and  $(p1, q1, q2)$  are different. Orientations of  $(p2, q2, p1)$  and  $(p2, q2, q1)$  are same



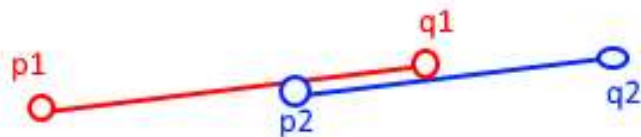
**Example:** Orientations of  $(p1, q1, p2)$  and  $(p1, q1, q2)$  are different. Orientations of  $(p2, q2, p1)$  and  $(p2, q2, q1)$  are same.

# How is Orientation useful here?

## 2. *Special Case*

- $(p1, q1, p2)$ ,  $(p1, q1, q2)$ ,  $(p2, q2, p1)$ , and  $(p2, q2, q1)$  are all collinear and
- the x-projections of  $(p1, q1)$  and  $(p2, q2)$  intersect
- the y-projections of  $(p1, q1)$  and  $(p2, q2)$  intersect

Examples:



**Example:** All points are collinear. The x-projections of  $(p1, q1)$  and  $(p2, q2)$  intersect. The y-projection of  $(p1, q1)$  and  $(p2, q2)$  also intersect



**Example:** All points are collinear. The x-projections of  $(p1, q1)$  and  $(p2, q2)$  do not intersect. The y-projection of  $(p1, q1)$  and  $(p2, q2)$  do not intersect

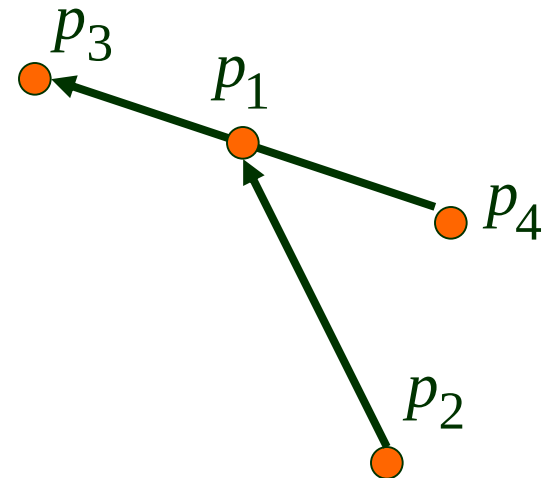
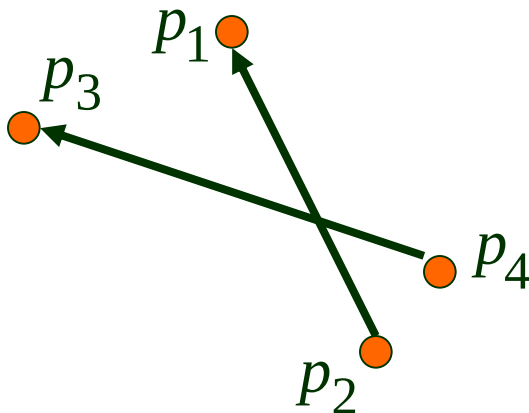


# Necessary and Sufficient Condition

Two line segments intersect iff **each** of the two pairs of cross products below have different signs (or one cross product in the pair is 0).

$$(p_1 - p_4) \times (p_3 - p_4) \text{ and } (p_2 - p_4) \times (p_3 - p_4)$$
$$(p_3 - p_2) \times (p_1 - p_2) \text{ and } (p_4 - p_2) \times (p_1 - p_2)$$

// the line through  $p_3 p_4$   
// intersects  $p_1 p_2$   
// the line through  $p_1 p_2$   
// intersects  $p_3 p_4$ .



Def: Cross Product (Next Slide)

## Problem:

You are given four points in 2D space:

- $P_1 = (1, 1)$
- $P_2 = (4, 4)$
- $P_3 = (1, 4)$
- $P_4 = (4, 1)$

Determine if the line segments  $P_1P_2$  and  $P_3P_4$  intersect.

- $P_1 = (1, 1)$
- $P_2 = (4, 4)$
- $P_3 = (1, 4)$
- $P_4 = (4, 1)$

## Solution:

### Step 1: Calculate the cross products

We need to calculate the cross products for each pair of vectors as described in the formula.

1. Calculate  $(P_1 - P_4) \times (P_3 - P_4)$

- $P_1 - P_4 = (1 - 4, 1 - 1) = (-3, 0)$
- $P_3 - P_4 = (1 - 4, 4 - 1) = (-3, 3)$
- Cross product:

$$(-3, 0) \times (-3, 3) = (-3)(3) - (0)(-3) = -9$$

2. Calculate  $(P_2 - P_4) \times (P_3 - P_4)$

- $P_2 - P_4 = (4 - 4, 4 - 1) = (0, 3)$
- $P_3 - P_4 = (-3, 3)$  (already calculated)
- Cross product:

$$(0, 3) \times (-3, 3) = (0)(3) - (3)(-3) = 9$$

- $P_1 = (1, 1)$
- $P_2 = (4, 4)$
- $P_3 = (1, 4)$
- $P_4 = (4, 1)$

**Solution:**

3. Calculate  $(P_3 - P_2) \times (P_1 - P_2)$

- $P_3 - P_2 = (1 - 4, 4 - 4) = (-3, 0)$
- $P_1 - P_2 = (1 - 4, 1 - 4) = (-3, -3)$
- Cross product:

$$(-3, 0) \times (-3, -3) = (-3)(-3) - (0)(-3) = 9$$

4. Calculate  $(P_4 - P_2) \times (P_1 - P_2)$

- $P_4 - P_2 = (4 - 4, 1 - 4) = (0, -3)$
- $P_1 - P_2 = (-3, -3)$
- Cross product:

$$(0, -3) \times (-3, -3) = (0)(-3) - (-3)(-3) = -9$$

- $P_1 = (1, 1)$
- $P_2 = (4, 4)$
- $P_3 = (1, 4)$
- $P_4 = (4, 1)$

**Solution:**

### Step 2: Check the signs

Now we check the signs of the cross products:

- $(P_1 - P_4) \times (P_3 - P_4) = -9$  and  $(P_2 - P_4) \times (P_3 - P_4) = 9 \rightarrow$  different signs.
- $(P_3 - P_2) \times (P_1 - P_2) = 9$  and  $(P_4 - P_2) \times (P_1 - P_2) = -9 \rightarrow$  different signs.

### Step 3: Conclusion

Since **both** pairs of cross products have different signs, the line segments  $P_1P_2$  and  $P_3P_4$  **do intersect**.

# Example: Collinear and Intersecting Points

## Example for Collinear and Intersecting Points:

Let's choose the following points that are collinear:

- $p(1, 1)$
- $q(3, 3)$
- $r(5, 5)$

Since the slope of the line formed by these points is the same, they are collinear. Now, we will check if point  $q(3, 3)$  lies on the line segment formed by  $p(1, 1)$  and  $r(5, 5)$ .

## Explanation:

In this case:

- $q.x = 3, p.x = 1, r.x = 5$
- $q.y = 3, p.y = 1, r.y = 5$

# Example: Collinear and Intersecting Points (cont...)

## Explanation:

In this case:

- $q.x = 3, p.x = 1, r.x = 5$
- $q.y = 3, p.y = 1, r.y = 5$

- $p(1, 1)$
- $q(3, 3)$
- $r(5, 5)$

The condition inside the `onSegment` function is checked as follows:

if  $(q.x \leq \max(p.x, r.x))$  and  $(q.x \geq \min(p.x, r.x))$  and  $(q.y \leq \max(p.y, r.y))$  and  $(q.y \geq \min(p.y, r.y))$

Substitute the values:

if  $(3 \leq 5)$  and  $(3 \geq 1)$  and  $(3 \leq 5)$  and  $(3 \geq 1)$

This evaluates to `true`, meaning that point  $q(3, 3)$  lies on the line segment  $pr$ .

Thus, for this set of points, the function would return `true`, confirming that  $q$  lies on the line segment  $pr$ .